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Tracing the Unseen: A Deep Dive into Mobility Behavior Exposure

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Abstract

Mobility data provides rich insights into human behavior, enabling applications in transportation, retail, and marketing, among others. However, it also poses significant privacy risks, as the unicity of individual mobility patterns may enable user identification within the dataset. Existing measures that aim to quantify the degree of unicity in users' characteristics, here referred to as *user exposure*, primarily focus on spatio-temporal unicity (e.g., *Uniqueness*), while overlooking broader behavioral traits that may also increase identification risks. To address this limitation, we have proposed MoBES (Mobility Behavior-based Exposure Score), a method for quantifying user exposure in mobility datasets. MoBES models user mobility behavior in a multidimensional behavioral space and estimates exposure based on behavioral distance. It also provides a continuous value score, enabling a more refined, nuanced analysis of user exposure. Building on our preliminary study of MoBES, this work considerably broadens and deepens the analysis of its design considerations for user exposure based on mobility behavior. To this end, we evaluate MoBES on two distinct Call Detail Record (CDR) datasets and analyze its behavior in a privacy-preserving setting where location data is perturbed using Differential Privacy (DP). Using two datasets enables a comprehensive comparative analysis, offering deeper insight into how behavioral patterns influence exposure levels. Our findings show that, although MoBES often aligns with traditional spatio-temporal measures, it also reveals some potential behavioral vulnerabilities that such measures fail to capture. Furthermore, we find that privacy protection mechanisms can effectively reduce MoBES exposure scores; however, the extent of this reduction varies across behavioral dimensions.

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This variation reveals that even under privacy protection, certain metrics could still be explored by potential adversarial methods.

1 Introduction

Human mobility, the act of moving from one place to another, is a fundamental aspect of daily life. As such, analyzing human mobility patterns can offer valuable insights into user behavior, supporting a wide range of applications across domains such as transportation, retail, and marketing. The widespread adoption of ubiquitous and pervasive computing technologies has also enabled faster and more informed mobility analyses [1], facilitating applications such as traffic forecasting, point-of-interest recommendations, and urban planning [2, 3].

However, mobility data can also compromise individual privacy by revealing sensitive personal information (e.g., home and work locations, visits to hospitals), or by exposing personal affiliations such as political or religious beliefs. Moreover, prior research has shown that users have highly unique mobility patterns that distinguish them from others in a given mobility dataset [4, 5], which can lead to their **identification**. For instance, De Montjoye *et al.* demonstrated that just four spatiotemporal points are sufficient to uniquely identify 95% of users in a large mobility dataset [6]. These findings highlight the notion of **user exposure**, defined as the degree to which an individual's observable mobility characteristics increase their unicity within a population as captured by a given dataset, thereby increasing their susceptibility to identification and related privacy risks. In this context, a user is considered exposed when their characteristics are quantifiably more distinguishable from those of the broader population.

One of the most widely used measures to estimate exposure in mobility data was proposed by De Montjoye [6]. Their method quantifies exposure by measuring the number of *unique* time-ordered sequences of visited locations extracted from a user's mobility trajectory. The more unique a user's movement patterns are, the higher the likelihood that the user can be uniquely identified. Although De Montjoye [6] was not the first to investigate unicity in mobility datasets, it was the first to formally quantify this concept, introducing a measure called *Uniqueness*. Therefore, throughout this paper, we use the term *Uniqueness* specifically to refer to the formulation proposed by De Montjoye [6]. As defined, *Uniqueness*, as well as related adaptations [7], estimates user exposure solely from the perspective of sequential visitation patterns. Furthermore, it is computationally expensive, as it requires evaluating temporal combinations of visited locations across all user trajectories. Finally, although *Uniqueness* is inherently a continuous measure, it is often used as a binary indicator (i.e., a user is either exposed or not), which limits its ability to capture varying degrees of exposure.

Instead, we argue that human mobility, as an expression of human behavior, is inherently multifaceted, thus requiring a multi-perspective analytical approach. Indeed, prior studies have characterized mobility using diverse metrics, each with its

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own goal, such as radius of gyration, diversity, regularity, stationarity, and entropy [8–11]. Going beyond simply counting the number of raw spatial-temporal sub-sequences (such as *Uniqueness*), these mobility metrics encapsulate processed insights into how individuals occupy space over time, capturing multiple facets of user mobility behavior. For example, the radius of gyration measures how confined a user's movements are, while diversity assesses how distinct the sub-sequences of visited locations are, i.e, a proxy to capture how often the user follows different spatial sub-trajectories.

Although these metrics may be correlated they offer complementary perspectives on one's mobility. For example, on one hand, broader displacements often suggest greater diversity in visited locations. On the other hand, diversity helps interpret the radius of gyration by revealing whether a user visits a variety of distinct locations in the users' usual area of confinement or travels between a few, but distant places, such as home and workplace. Thus, the joint use of multiple metrics naturally defines a multidimensional behavioral space, each dimension reflecting one mobility metric, where the mobility behavior of a user can be represented. *Such representation allows for the assessment of new forms of user exposure in mobility patterns, namely, exposure due to the unicity of mobility behavior*, here referred to as **behavior exposure**.

Prior studies have explored multiple behavioral metrics in the context of user exposure [12, 13]. However, these works do not focus directly on the identification task itself, but rather on the risk of re-identification. In this setting, users' prior behavioral information is leveraged to re-identify them under different scenarios. In both cases, the problem is framed as a supervised binary classification task. Consequently, although these approaches can be used to estimate exposure, they face the fundamental challenge of the frequent absence of ground truth for training data construction. To generate labels, these methods rely on simulating specific adversarial models, defining exposure risk as the probability that a user will be **re-identified** in the dataset according to a predefined threat model. While these approaches incorporate multiple perspectives of mobility behavior, they present key limitations: (i) they are inherently tied to the specific simulated adversarial models, which may not fully capture realistic or unforeseen attack strategies; and (ii) the simulated attacks can be computationally demanding, particularly when combinations of visitation patterns are considered [12]. Thus, although such techniques provide a way to quantify user exposure, *Uniqueness* remains one of the most widely adopted approaches in the literature.

Recognizing the limitations of prior methods, we have introduced MoBES, Mobility Behavioral Exposure Score [14], that leverages a set of mobility metrics to quantify exposure. It does so by modeling the mobility behavior of a user as a vector of metric values in a multidimensional metric-based behavioral space. MoBES quantifies a user's exposure based on the average distance between its vector representation and the vectors of its k nearest neighbors in the defined space. In broad terms, users who are close to their neighbors are *protected* by them, as they cannot be easily distinguished in the crowd. Users who are farther from their neighbors, in turn, have a greater exposure, as they exhibit unique behavioral patterns. As defined, MoBES is not tied to any specific adversarial model but instead captures fundamental aspects of user behavior. It also offers a continuous value score that allows for a more refined and

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nuanced analysis of user exposure. Overall, MoBES contributes to the state-of-the-art by providing a novel, behavior-driven assessment of user exposure in mobility data.

In our preliminary study [14], MoBES was evaluated on a single network-sourced cellular dataset collected from Shenzhen, China [15]. However, different datasets may exhibit very distinct mobility patterns [9], which can in turn significantly impact user exposure assessment. In this work, we aim to build on our prior study by providing a much more comprehensive investigation of user exposure based on mobility behavior using MoBES. We do so by expanding our prior study in two directions.

First, we bring a second mobility dataset to our investigation: A pseudonymized network-sourced cellular dataset gathered by a Telecom operator in the metropolitan area of Shanghai, China. Both Shenzhen and Shanghai datasets exhibit very different spatial and temporal properties and diverse population characteristics [9]. We investigate properties of `MoBES` for users in both datasets, comparing it against the widely used *Uniqueness* measure. We show how `MoBES` contributes to the state-of-the-art by offering a complementary analysis of mobility behavioral exposure. *To support reproducibility, we make the MoBES code publicly available [16].*

Second, we analyze user exposure according to `MoBES` in case the mobility data is first protected against privacy violations by applying the state-of-the-art centralized Differential Privacy (DP) method [17]. DP is a mathematical framework for ensuring data privacy, which works by carefully adding calibrated noise to the input data, making it difficult to identify individuals' data within the dataset. Our present goal is to assess the extent to which the use of DP to protect spatial information impacts user exposure as measured by `MoBES`. To our knowledge, this is the first effort to quantify user behavioral exposure in a privacy-preserving setting.

In sum, this work builds on our preliminary study [14] by offering the following novel contributions:

- An evaluation of `MoBES` and *Uniqueness* on two pseudonymized network-sourced cellular datasets, which have very distinctive properties in terms of user mobility patterns, enriching the discussion and evaluations of both methods. Our analysis reveals users who, although not unique in the sequences of places they visit, exhibit mobility behavioral patterns that are quite distinct from others, making them more exposed. The use of two datasets enables a comprehensive analysis, offering deeper insights into how different behavioral patterns influence user exposure.
- An investigation of the impact of using Differential Privacy (DP), to protect location data, on user exposure assessment via `MoBES` in both datasets. Our results reveal two key insights: (*i*) DP does affect user behavior metrics, but the extent varies across metrics, some are significantly altered even with low noise levels. (*ii*) As noise increases, MoBES scores decline accordingly, indicating that user exposure decreases with higher levels of privacy protection. This demonstrates MoBES's potential as a diagnostic tool for assessing and balancing the trade-off between data utility and user privacy.

The remainder of this article is organized as follows. We start by discussing background for our work (Section 2) and formalizing the design of `MoBES` (Section 3). We then describe the two real-world datasets used in our study (Section 4), discuss how

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MoBES can be instantiated to analyze user exposure in them (Section 5), and compare MoBES with the widely used *Uniqueness* measure (Section 6). Next, we proceed to discuss the use of DP on both datasets and its impacts on MoBES and *Uniqueness* (Section 7). We finish with a discussion on methodological considerations regarding the design of Mobes (Section 8), followed by conclusions and directions for future work (Section 9.

2 Scope and motivation

This section offers a foundational background on user exposure in mobility data. It begins by addressing privacy risks and protection mechanisms in mobility data, and then focuses on quantifying user exposure, the central topic of this paper.

2.1 Privacy risks in mobility data

Privacy risks associated with mobility data have been extensively discussed across different domains. It is well established that the exposure of seemingly innocuous personal data can enable malicious identification. Rocher *et al.*, for example, demonstrated that 99.98% of U.S. residents could be uniquely identified using only 15 demographic attributes [18]. In the context of mobility, this concern is exacerbated by the ubiquity of location-tracking technologies embedded in smartphones and other personal devices.

Although location data holds great promise for societal and commercial applications [19], it can also pose severe privacy threats, and the main key to different privacy threats lies in the user exposure, i.e., the degree to which an individual's observable mobility characteristics increase their **unicity** within a population (as captured in a given dataset), leading to an increase in the susceptibility to identification. Indeed, Zang and Bolot showed that knowing a user's top three visited locations was sufficient to uniquely identify 50% of a sample of 25 million users [4]. Similarly, De Montjoye *et al.* found that just four time-ordered spatial points could uniquely identify 95% of 1.5 million individuals [6]. More recently, Farzanehfar *et al.* reported identification rates as high as 93% in nationwide datasets covering over 60 million users [5]. These findings reveal two essential insights: *(i) minimal knowledge of a user's trajectory can suffice for identification, and (ii) this risk persists even in large-scale datasets.* At the core of user exposure issues lies the inherent unicity of human mobility, shaped by geographical, temporal, contextual, and social factors [1], which are the characteristics that directly drive behaviors in mobility.

To mitigate such risks, several privacy-preserving techniques have been adapted to mobility data, aiming to hide the mobility fingerprints of users. These include K-Anonymity, Differential Privacy (DP), and the use of Generative Adversarial Networks (GANs), which were initially proposed in other contexts and were adapted to mobility data. K-anonymity, in brief, guarantees that at least other k users share similar attributes. In the context of mobility data, Gramaglia and Fiore [20], proposed Glove, a technique that guarantees that at least k other users will share similar trajectories. In sum, Glove iteratively groups spatial-temporal points in k clusters, making the users indistinguishable. In contrast, Differential Privacy [17] works by applying a controlled

noise over the data. It has been directly applied to mobility data by adding such noise to geo-location data, ensuring the obfuscation of users' exact location [21, 22]. Lastly, GANs were initially proposed in the context of computer vision [23] with the intent of generating synthetic images that were indistinguishable from the original training data. Given the efficiency of these techniques in several contexts [24], it was quickly adapted to mobility data, as in LSTM-TrajGAN [25], which adapts the architecture of a GAN by also adding an LSTM (Long-Short Term Memory) layer. Such a layer allowed the GAN architecture to understand the temporal patterns in a trajectory and consequently generate high-quality synthetic data. Nonetheless, while effective to some degree, *these methods face limitations, notably high computational costs and the difficulty in quantifying the level of protection achieved.* Additionally, many of these techniques were initially designed for other data types and *may not fully capture the intricacies of mobility data.*

2.2 User exposure quantification

A key direction in mobility privacy research is the quantification of user exposure. One foundational measure is *Uniqueness*, which quantifies how distinguishable a trajectory is within a dataset [6]. *Uniqueness* typically evaluates all ordered subsequences of spatiotemporal points within a time window of duration t_w. A *Uniqueness* score of one implies that a user had at least one unique subsequence, indicating full exposure at that point in time. However, *this method is computationally expensive, prompting approximations that consider only ordered time points* [7].

In contrast to measures like *Uniqueness* that operate directly on raw trajectory data, other methods first process these trajectories to extract mobility metrics, working on higher-level abstractions that capture behavioral patterns, which are then used to estimate user exposure [12, 13, 26]. For example, Rodríguez-Hart *et al.* [26] introduced an entropy-based metric, rooted in the work of Song *et al.* [10], which links predictability to user exposure. In this view, more predictable movement patterns, such as frequent visits to a small set of locations, indicate higher exposure. However, *this entropy-centric view offers a limited, one-dimensional perspective.*

To address this, Pellungrini *et al.* [12] and Naretto *et al.* [13] proposed supervised machine learning approaches that incorporate multiple mobility metrics. These methods rely on labeled data to model exposure as the output of adversarial re-identification models. Both works extract a common set of mobility metrics from user trajectories but differ in their objectives: Pellungrini et al. assess the probability of user re-identification across eight adversarial models, while Naretto et al. proposed EXPERT, a system that provides a binary classification, re-identified or not, focused specifically on location sequence attacks. Although more comprehensive, *these models are tightly coupled to specific threat models and often demand high computational resources.*

Despite notable contributions, the aforementioned existing methods for exposure quantification in mobility data present several critical limitations, notably:

Binary assessments: Many measures, such as *Uniqueness* and EXPERT, treat exposure as a binary variable (e.g., unique vs. not unique; high vs. low risk) [6, 13]. This

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oversimplification hampers the application of fine-grained protection strategies. In contrast, continuous measures (as in [12]) support more nuanced evaluations and targeted interventions.

Adversarial model dependence: Supervised approaches typically define exposure based on specific adversarial models, which may not reflect real-world attack scenarios [12, 13]. This dependency limits generalizability and may lead to a false sense of security.

High computational cost: Measures like *Uniqueness* require exhaustive enumeration of spatiotemporal subsequences, becoming prohibitively expensive for large or complex datasets. Even machine learning-based models incur significant computational overhead. For instance, Pellungrini *et al.* report needing over a week to compute exposure for a dataset with one million users [12].

Limited perspective: Some methods, especially *Uniqueness*, focus narrowly on specific patterns (e.g., location sequences) and fail to capture broader behavioral traits or contextual factors. In contrast, multi-metric models can accommodate a more holistic view of mobility behavior and adapt to varying data types and domains.

To address the limitations outlined above, we introduced MoBES, the Mobility Behavior-based user Exposure Score [14], a measure designed *to quantify user exposure in mobility datasets based on how behaviorally distinguishable a user is from their peers.* While our initial evaluation of MoBES was limited to a single dataset, *the present work significantly extends that effort by exploring exposure quantification across a much broader range of scenarios.* Specifically, we assess and compare the performance of MoBES and the widely adopted *Uniqueness* measure across multiple mobility datasets and in setups where privacy-preserving mechanisms are employed to protect location data.

3 Quantifying user exposure with MoBES

MoBES was designed to quantify the *exposure of a user* in a mobility dataset in terms of how *unique* the mobility behavior of the user is with respect to the behavior of the population represented in the dataset. To that end, MoBES leverages a set of mobility metrics to build a multidimensional behavioral space, where each user is represented as a vector of metric values. MoBES design is driven by the hypothesis: *users who are far from their closest neighbors in the defined multidimensional metric-based space exhibit very distinguishable behavioral signatures, being more exposed.* On the contrary, *users who are close to their nearest neighbors in the multidimensional metric-based space tend to share similar behavioral patterns, thereby reducing their susceptibility of being identified.* We further elaborate on MoBES design and the defined behavioral space next.

3.1 MoBES formalization

Consider a human mobility dataset comprising a set $\mathcal{U}$ of n users. Each user $i \in \mathcal{U}$ is associated with a mobility trajectory, representing a sequence of tuples (t, l) where l corresponds to a geographical location (i.e., a longitude-latitude pair) visited by i at time t. We define a set $\mathcal{M}$ of m mobility metrics, where each metric can be

computed from an individual user's trajectory, capturing some processed information regarding the user's mobility patterns. MoBES design is general and *does not specificy any particular set of metrics.* Ideally, set $\mathcal{M}$ should include various metrics which, while capturing different perspectives of one's mobility patterns, collectively offer a more holistic representation of user mobility behavior. We defer to Section 3.2 a more detailed discussion of the metrics used in this study.

Given user i's trajectory, we compute $\mathcal{M}_i = \{m_1^i, m_2^i, \ldots, m_r^i, \ldots, m_m^i\} \in \mathbb{R}^+$, where m_r^i denotes the value of metric r for user i. This results in a representation of user i in an m-dimensional behavioral (metric-based) space, where each dimension corresponds to a specific mobility metric in $\mathcal{M}$. The space defined by $\mathcal{M}$ thus models user mobility behavior at a higher level of abstraction, capturing patterns and characteristics beyond raw trajectories. We refer to the vector $\mathcal{M}_i$ as the **mobility signature** of user i, as it uniquely characterizes i's behavioral profile within this metric space. This behavioral representation forms the foundation for computing the Mobility Behavior-based user Exposure Score (MoBES).

To quantify the exposure underlying a user's mobility signature, MoBES assesses *how behaviorally distinct the user is relative to others in the dataset.* This is achieved by computing a **relative distance** between users in the defined behavioral space. Specifically, MoBES compares the mobility signature of a target user i to that of another user j, referred to as a neighbor of i, by computing the average relative absolute difference across all behavioral metrics. That is, for each metric, it computes the absolute difference between the values for users i and j, and divides it by the corresponding value for user i, and then averages the resulting values across all metrics. This chosen distance metric, i.e., the L1 norm of the relative differences provides an interpretable measure of behavioral deviation, capturing the average relative difference in each mobility metric between users. The resulting value reflects how dissimilar user i's behavior is from that of neighbor j, on average, with higher values indicating greater behavioral distinctiveness, and hence, higher exposure (cf. Section 8 for a more detailed discussion).

Formally, given a target user i and a neighbor j ($i, j \in \mathcal{U}$), we define the **distance from i to j** in the behavioral space as:

$$D_{i,j} = \frac{1}{m} \sum_{r=1}^{m} \left| \frac{m_r^i - m_r^j}{m_r^i} \right|, \tag{1}$$

where m_r^i and m_r^j denote the values of the rth metric for users i and j, respectively. In a sense, this distance quantifies the extent to which user i would need to adjust its behavior on average, across all dimensions, to resemble user j. Intuitively, the more distinct i is, the greater the adjustment required to *"blend in"* with a neighbor, reflecting a higher degree of behavioral exposure.

It is important to note that the values $D_{i,j}$ are relative to the target user i, being thus asymmetric, i.e., $D_{i,j} \neq D_{j,i}$. Moreover, as defined in Equation 1, all metrics are treated with equal importance. Yet, one could assign different weights to prioritize specific behavioral dimensions, which we leave as an avenue for future exploration.

Building on Equation 1, we quantify the exposure of a target user i based on i's relative distance to other users in the defined behavioral space. Specifically, we

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compute $MoBES_i$, the value of MoBES for user i, as the *average distance from user i to its k-nearest neighbors in the defined behavioral space*. In other words, let NN_i^k be the set of the k nearest neighbors of i, identified by calculating the distances between i and all other users in the dataset using Equation 1. The **exposure** of user i is defined as:

$$MoBES_i = \frac{\sum_{j \in NN_i^k} D_{i,j}}{k} \tag{2}$$

This formulation captures how behaviorally isolated or distinguishable a user is relative to its local neighborhood. The parameter k is configurable, allowing the exposure score to be tailored to different sensitivity levels: smaller values of k yield a stricter notion of exposure (focusing on fewer closer neighbors), while larger values produce a more relaxed assessment by considering a broader neighborhood.

Note that, as defined, *the score of MoBES produces a continuous-valued estimate of exposure*, enabling fine-grained analysis of the users' exposure. This represents a significantly richer framework compared to the binary classification adopted by other measures [6, 13]. Depending on the context of the application, one may define multiple exposure thresholds to partition the MoBES values into distinct exposure categories. Both the number of classes and the choice of threshold values defining such classes are context-dependent: *a larger number of classes allows for more nuanced and fine-grained analyses, while higher thresholds correspond to less strict definitions of exposure, identifying only users with more pronounced deviations in behavior.*

In this article, to facilitate the analysis and align with previous efforts, we adopt a binary classification scheme defining as exposed users according to a single threshold given by parameter τ. A user i is deemed exposed if $MoBES_i \geq \tau$. The users that satisfy such a condition are part of the **set of exposed users**, denoted by $MoBES^*$. Formally, $MoBES^* = \{i, \forall i \in \mathcal{U}$ such that $MoBES_i \geq \tau\}$. As discussed above, the choice of τ is context-dependent.

3.2 Behavioral space representation

To construct the m-dimensional behavioral space, we use a set of existing mobility metrics that capture different aspects of user movement. Collectively, these metrics enable a comprehensive representation of mobility patterns, forming the basis for computing MoBES. While we do not propose new metrics, our goal is to capture users' multi-dimensional mobility behavior through a single, continuous exposure score.

Specifically, we select 13 well-established state-of-the-art mobility metrics frequently used in the literature to characterize user behavior [8, 27, 28]. These metrics are organized into three broad categories, namely:

- *Spatial metrics:* describe the geographic extent and dispersion of user movements across physical space;
- *Temporal metrics:* capture behavioral patterns derived from timing and duration of visits; and
- *Structural metrics:* reflect sequential patterns of visits, incorporating both spatial and temporal dimensions.

Table 1 presents the selected metrics along with a brief description of the aspect of mobility behavior each captures. Collectively, these metrics account for spatial-temporal regularities, individual preferences, and behavioral variability, enabling the identification of diverse patterns of exposure.

At this point, we highlight two important points:

- Metric selection is **inherently context-dependent**. For this study, we select metrics that can be computed directly from the available datasets, introduced in Section 3.3, and deliberately prioritized a broader behavioral space to capture multiple facets of mobility. Yet, the MoBES framework is intentionally generic and flexible. New or alternative metrics can be incorporated as additional dimensions, enabling refined, context-specific modeling as needed. In particular, contextual metrics, such as Point of Interest-based metrics [29], could also be incorporated when derivable from the available data, representing a promising direction for future work.
- Correlations may exist. While some correlation between metrics may exist, we here choose to keep all selected metrics, as they capture complementary perspectives of mobility behavior, enabling a more meaningful assessment of exposure, as discussed in Section 3.3. Collectively, they provide a richer and more nuanced understanding of individual mobility patterns [30] and their associated exposure compared to prior work [6].

Finally, since the adopted distance metric (Equation 1) is based on *relative differences*, the resulting values can be used directly for exposure assessment without requiring normalization of the metric values. This property further enhances the interpretability and consistency of MoBES across different metrics and datasets, as discussed in Section 8.

3.3 MoBES overview: Core strengths and integrative design

MoBES introduces a novel perspective for assessing and interpreting user exposure grounded in mobility behavior. It constitutes an **original and integrative** measure that represents individual behavioral patterns in an m-dimensional metric-based space, capturing multiple facets of user mobility. In doing so, it identifies core behavioral characteristics that may lead to exposure, while remaining **non-dependent on supervised techniques**. MoBES is both **flexible and adaptable**, supporting various instantiations based on different sets of metric and parameter configurations to suit diverse application contexts. Owing to its reliance on computationally simple operations, MoBES enables a **scalable and tunable** assessment of exposure. Also, by generating a continuous exposure score instead of binary classifications, it offers a more **nuanced and interpretable** understanding of individual exposure.

A notable strength of the framework lies in its **parameter flexibility**, particularly through the use of k and τ, which allow for adjustable levels of analytical precision concerning **crowd-blending** and **exposure sensitivity**, respectively:

- The **parameter** k controls the size of the neighbor group to which a user is compared, effectively determining the desired level of *crowd-blending*. A small k (e.g., $k = 1$) offers a localized view, comparing the user only to its "nearest" neighbors in

Table 1: Mobility metrics used to construct the MoBES behavioral space, grouped into spatial, temporal, and structural dimensions. Together, these metrics capture complementary aspects of mobility behavior, enabling a multidimensional representation of users and a richer assessment of behavioral exposure.

Type	Metric	Notation	Description
Spatial	Radius of gyration [11]	RoG	Captures the degree of spatial confinement in a user's mobility. Lower values indicate more restricted movement patterns. It is computed as the average distance from each visited location to the trajectory's center of mass (i.e., location corresponding to the average latitude and average longitude of all visited locations).
	2-Radius of gyration [31]	2kRoG	Captures the extent of displacement relative to the user's two most dominant routine locations (typically home and work), which explains the "2" in the metric's name. It reflects how much the user's mobility deviates from their core routine anchors.
	Maximum displacement [32]	MDIST	Largest spatial displacement observed between any two consecutive locations in the user's trajectory. It reflects the most significant single-step movement during the observation period.
	Average and standard deviation of jump lengths [11]	AJL and SJL	Average and standard deviation of the distance between consecutive displacements made by an user.
	Number of visits [8]	NoV	Number of records in the user's trajectory. Captures the amount of information regarding the user mobility patterns available in the data.
	Number of unique locations [11]	NUL	Captures the diversification of user mobility in terms of distinct locations visited during the observation period.
Temporal	Average and standard deviation of waiting times [10]	AWT and SWT	Average and standard deviation of the time spent in each location, inferred from the available data.
Structural	Diversity [30]	DIV	Captures how diverse a user's visitation patterns is in terms of the distinct subsequences of visited locations.
	Regularity [27]	REG	Captures if a user is prone to return to previously visited locations.
	Stationarity [33]	STAT	Quantifies if an individual stays continuously in the same location.
	Real Entropy [10]	ENT	Captures the uncertainty in the user mobility patterns. It builds upon the visitation frequency, the order in which the locations were visited, and the time spent at each location.

the behavioral space, with behavioral similarity influenced by the degree of proximity (i.e., the distances between them). On the other hand, bigger values of k define a more structured neighborhood, which will have little impact on the users' exposure in scenarios of behavioral drift by one of the peers.

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- The **threshold** τ defines the minimum behavioral distance required for a user to be considered exposed, enabling the adjustment of *exposure sensitivity*. Small values of τ reflect a stricter or more cautious exposure policy, i.e., even minor behavioral distances trigger exposure. Instead, high τ values reflect a more tolerant or conservative exposure threshold, indicating users must be clearly distinct from others to be considered exposed.

One of MoBES key advantages lies in its **high interpretability and observability**, i.e., respectively, the abilities to (i) explain what it is doing and why; and (ii) the ability to infer internal states (exposure causes) from external measurements (behavioral metrics). First, representing user mobility in an m-dimensional behavioral space enables a fine-grained understanding of which behavioral traits contribute most to a user's exposure. Second, the distance function of Equation 1 is easy to interpret, offering a continuous, intuitive measure of divergence from nearby neighbors along each behavioral axis. Finally, one can *quantify the relative contribution of each metric to the user's overall distance to its k nearest neighbors*. In other words, for a given user i and a behavioral metric r, the contribution of that metric to the final MoBES score of i, named *metric importance*, is formally captured by Equation 3:

$$I_{i,r} = \frac{\sum_{j \in NN_i^k} \left| \frac{m_r^i - m_r^j}{m_r^i} \right|}{m \times \sum_{j \in NN_i^k} D_{i,j}} \tag{3}$$

Here, a higher value of $I_{i,r}$ indicates that the metric r plays a stronger role in differentiating user i from its k nearest neighbors, revealing which behavioral dimensions most influence exposure. This insight allows users (or systems acting on their behalf) *to identify actionable patterns and assess how targeted behavioral adjustments might reduce exposure or achieve greater similarity with the surrounding population.*

As a consequence, MoBES opens the door for designing **tailored protection mechanisms**, aligned with (i) the user's most distinguishing behavioral traits and (ii) those users most vulnerable to exposure. This allows for selective and context-aware interventions, avoiding *one-size-fits-all* protections and, instead, enabling more precise, efficient, and sustainable privacy-preserving strategies, even for users at relatively lower susceptibility of being identified.

In the following sections, we present results from a specific instantiation of MoBES, where user exposure is estimated based on the set of behavioral metrics introduced in Section 3.2. Before discussing these results, we first describe the datasets used in our study.

4 Mobility datasets

In this section, we first present an overview of the two real-world datasets used to illustrate how MoBES works (Section 4.1). We then present a brief characterization of each dataset with respect to the mobility metrics used to define MoBES' m-dimensional behavioral space (Section 4.2).

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4.1 Dataset overview and pre-processing

Our study relies on two pseudononymized network-sourced datasets, i.e., Call Detail Records (CDRs), capturing mobile communication activities (calls, SMS) within Telecom network operators. CDRs typically cover large populations and broad geographical areas (e.g., cities), offering valuable insights into mobility behaviors. The datasets describe communication events occurring in specific regions of Shenzhen [34, 35] and Shanghai cities, in China. While Shenzhen is publicly available [35] and regroups information from several telecom operators' networks in the city, the Shanghai dataset's access is restricted and relates to a major cellular operator's network. The selection of these datasets was guided by their relevance in the literature [9, 14, 15, 30, 36] and their characteristics, as discussed below, which offer valuable complementary scenarios for evaluating user exposure. For the sake of simplicity, throughout the rest of this article, we refer to each dataset by the name of the city from which it was collected, namely *Shanghai* or *Shenzhen*.

One key distinction between the two datasets is their spatio-temporal granularity. Unlike traditional CDRs, and for privacy compliance reasons, the Shanghai dataset was preprocessed by the data provider before sharing, to provide individual trajectories with a temporal resolution of approximately one hour [30, 37]. Accordingly, the locations in the Shanghai dataset represent each user's hour-centroid within a 200-meter radius and do not include the coverage area of each tower. By contrast, the Shenzhen dataset is a *standard event-based CDR dataset*, recording a user's cell tower location whenever a communication event, such as a call or SMS, occurs. Consequently, the Shenzhen dataset exhibits greater temporal and spatial sparsity than the Shanghai dataset, thereby affecting the mobility patterns captured in each dataset.

To ensure data quality and to avoid comparison biases when extracting user mobility behavior, each dataset underwent the following steps as part of our own pre-processing pipeline:

- **Time discretization:** Users with multiple records within the same hour and day had their geographical location averaged to ensure at most one record per user per hour, reducing redundancy and bias in metric extraction, notably temporal metrics.
- **Tessellation:** Each geo-coordinate was assigned to a unique quadratic grid of 200m $\times$ 200m, addressing spatial sparsity, and allowing the computation of metrics such as Number of Visits (NoV) and Number of Unique Locations (NUL).
- **Filtering:** Users with fewer than 10 active days or 100 records were removed to eliminate outliers with distorted behavior.
- **Completion:** When necessary and for short intervals, trajectory completion is applied to mitigate the common temporal sparsity in CDRs (i.e., dataset gaps corresponding to periods when phones were not communicating), without biasing results analysis [30, 36, 38, 39]. Specifically, we first identify two types of grids for each user, namely *home* and *work*, as the most frequently visited locations during *nighttime* and *daytime*, respectively. Then, for each hour within the time intervals $[2am, 6am]$, $[10am, 11am]$, and $[2pm, 4pm]$, if a record is missing, a new one is added using the most frequent identified location with the corresponding grid type: i.e.,

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Table 2: Summary statistics of the Shanghai and Shenzhen datasets, including user-level and visit-level mobility metrics (95% confidence intervals are within ± 0.01). Shanghai contains a larger user population and exhibits more visits and visited locations per user, whereas Shenzhen presents greater sparsity and a larger number of unique locations overall. These datasets highlight the distinct mobility scenarios used to evaluate our proposal.

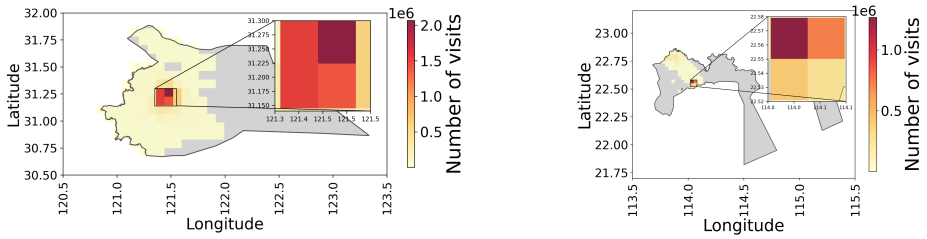
Metric	**Shanghai**	**Shenzhen**
Total number of users	58502	31859
Total number of days	10	15
Total number of unique locations	6064	8669
Avg. number of unique locations per user	12.4	9.67
Avg. number of unique locations per user per day	3.37	2.8
Avg. number of records/visits per user	180.37	136.01
Avg. number of records/visits per user per day	18.03	9.25

the users' identified grid-home for $[2am, 6am]$ and the grid-work for $[10am, 11am]$ and $[2pm, 4pm]$. As such, completion promotes temporal continuity over very short intervals under controlled assumptions, ensuring fair and consistent comparisons across datasets.

To further address the temporal sparsity of the Shenzhen dataset, we retained only the first 15 consecutive days, which correspond to the period with the largest number of active users. This filtering step reduces potential bias in the temporal metrics and prevents overestimation of users' exposure scores. Table 2 presents key statistics of the pre-processed datasets, highlighting their main differences.

As shown in the table, the Shanghai dataset includes almost twice as many users as Shenzhen, but shorter temporal coverage (10 versus 15 days). Moreover, though Shenzhen has a larger number of unique locations, Shanghai users tend to visit, on average, more locations both in total and per day, considering all visits and unique places. These patterns align with each city's spatial dispersion, as captured by the corresponding dataset, which is illustrated in Figure 1. The figure shows the spatial distribution maps of both cities. These maps reveal that Shenzhen's spatial distribution is significantly more constrained than Shanghai's, reflecting the fact the area of observation of the city of Shanghai is larger.

We highlight that, in our evaluation, we chose to apply MoBES at the city level, leveraging all available data for each city to estimate user exposure. However, it could potentially be applied at finer or coarser geographical granularities, such as at neighborhood level. Similarly, we chose to compute MoBES over the full temporal span of each dataset. Yet, it could be recomputed periodically or after some time period (e.g., to accommodate user behavior shifts that may impact user exposure). We leave the exploration of different spatial granularities and temporal recomputation strategies for future work.



(a) Shanghai (b) Shenzhen

Fig. 1: Spatial distributions of the cities of Shanghai and Shenzhen. Shanghai covers a broader geographical area and exhibits greater spatial dispersion, while Shenzhen presents a more spatially constrained mobility landscape. The zoom highlights zones with the highest numbers of visits. These differences contribute to the distinct exposure patterns observed across the two datasets.

4.2 Characterization of mobility metrics

In this section, we offer a characterization of the Shanghai and Shenzhen datasets using the 13 mobility metrics that define the m-dimensional behavioral space of MoBES (Section 3.2). This overview provides context for interpreting the user exposure results presented later. Figure 2 shows the distribution of each metric for both datasets.

We begin with the spatial metrics. Due to its larger geographic area, Shanghai exhibits consistently higher spatial dispersion than Shenzhen across RoG, 2kRoG, MDIST, AJL, and SJL. In contrast, Shenzhen users display more geographically confined mobility patterns, reflected in shorter travel distances and smaller radius of Gyration. Additionally, both the Number of Visits (NoV) and the Number of Unique Locations (NUL) in Shenzhen present distributions with longer tails, indicating the presence of a small group of highly mobile users. Such behavioral outliers may become particularly relevant when exposure is assessed across multiple dimensions.

Regarding the temporal metrics (AWT and SWT), Shanghai's distributions are more concentrated, whereas Shenzhen shows greater variability. This difference stems from the data collection protocols: Shenzhen records are event-driven, leading to heterogeneous sampling frequencies, while Shanghai data (after completion) follow a regular hourly interval. As a result, Shanghai exhibits more stable waiting time statistics.

Finally, we analyze the structural metrics (ENT, STAT, REG, DIV), which integrate spatial and temporal aspects. Most structural measures show similar patterns across datasets, except for stationarity (STAT), which is higher in Shanghai, suggesting longer stays at the same locations. Despite Shanghai users visiting more distinct places, diversity does not increase proportionally, likely reflecting more routine mobility patterns.

Overall, the two datasets display distinct spatial, temporal, and structural properties, resulting in different mobility behaviors. Notwithstanding, this also highlights the context-dependency in the selection of behavioral metrics, where some metrics will be

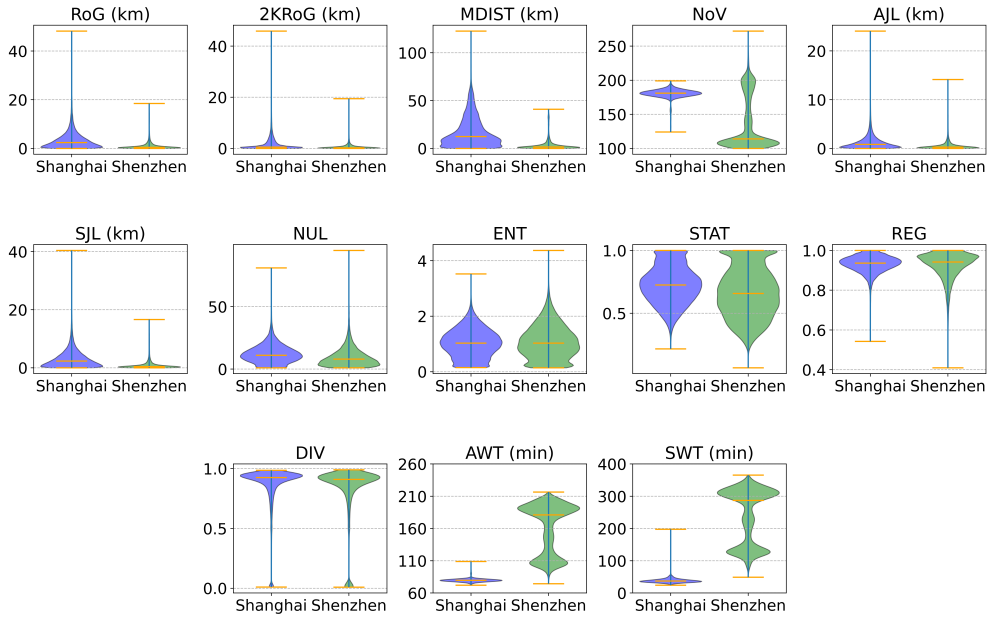


Fig. 2: Distributions of mobility metrics that compose the behavioral space for the datasets of Shanghai and Shenzhen. The two cities exhibit markedly different spatial, temporal, and structural mobility characteristics, highlighting that user exposure must be assessed from multiple behavioral dimensions rather than from a single mobility feature.

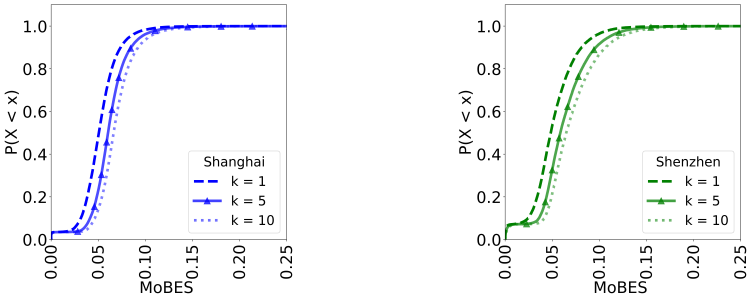
more distinguishable than others according to the characteristics of the city. Nonetheless, to avoid any bias when comparing the two scenarios, we selected the same set of metrics to evaluate both cities. In the next section, we examine how these differences translate into variations in user exposure.

5 MoBES in action

In this section, we present and discuss the nuances of user exposure in both the Shanghai and Shenzhen datasets. We begin by explaining how MoBES was parameterized, followed by an analysis of the results.

5.1 Parameterization

As presented in Section 3, MoBES relies on two main parameters: k and τ. The parameter k defines the number of nearest neighbors considered when computing a target user's average behavioral distance, thereby determining the degree of **crowd-blending** of a user. The parameter τ serves as a threshold for identifying exposed users, i.e. those whose distance to the nearest neighbors is greater than or equal to τ, meaning their score exceeds the acceptable **exposure sensitivity** threshold.



(a) Shanghai (b) Shenzhen

Fig. 3: Cumulative distributions of MoBES for increasing values of the neighborhood parameter k={1,5,10} in Shanghai (a) and Shenzhen (b). Larger values of k systematically increase exposure scores by incorporating more distant neighbors into the behavioral comparison. Across all configurations, Shenzhen presents a slightly heavier upper tail, indicating a larger proportion of highly exposed users.

In the following, we analyze how variations in these parameters influence MoBES performance.

We begin by examining the influence of the parameter k, which, as defined, controls the desired level of *crowd-blending* of a user (cf. Section 3.3). To this end, we evaluate MoBES using three values, 1, 5, and 10, chosen to represent different scenarios. A small value such as $k = 1$ reflects a minimal protection setting, where a target user may be considered protected if at least one similar neighbor exists. However, this protection is fragile: if that neighbor changes behavior, the user exposure may increase. In contrast, larger values ($k = 5$ or $k = 10$) offer more stable and robust estimates of user exposure, as they rely on a broader and more consistent behavioral neighborhood in which a user can blend.

Figure 3 presents the distributions of MoBES for different choices of k across both datasets. As k increases, the distribution shifts rightward, reflecting higher MoBES values. This is expected, as a larger k includes more distant neighbors in the m-dimensional behavioral space, thereby raising the average distance and, consequently, the exposure estimates. Across all values of k, the Shenzhen dataset consistently yields a MoBES distribution with a slightly longer tail, suggesting a larger fraction of users with a greater exposure. For example, by analyzing the 80th percentile of each distribution, we find that the top 20% of users in Shenzhen with the highest exposure have MoBES values of at least 0.07 for $k = 1$ (at least 0.08 for $k = 5$). For Shanghai, the same top 20% of users have MoBES values of at least 0.06 for $k = 1$ (0.07 for $k = 5$).

We now turn our attention to the role of the threshold parameter τ. As defined, τ represents a level of *exposure sensitivity*, i.e., a permissible degree of behavioral deviation, beyond which a user is classified as exposed. Selecting an appropriate τ is critical, as it directly affects the sensitivity of exposure detection. Obviously, the lower the value of τ, the greater the number of users considered exposed. Thus, the choice of τ should reflect the desire for stricter (lower τ) or more flexible (higher τ)

notions of exposure. In Section 6, we compare user exposure according to MoBES and Uniqueness using three values of τ: $\{0.05, 0.10, 0.15\}$. At $\tau = 0.05$, even minor deviations from the k nearest neighbors are enough to flag a user as exposed, capturing subtle behavioral differences. In contrast, a higher threshold like $\tau = 0.15$ identifies only users with more pronounced deviations. Before analyzing the differences between MoBES and Uniqueness, we delve deeper into user exposure assessed by MoBES by characterizing the typical mobility behavior of users with various values of MoBES. These results are discussed next.

5.2 User exposure characterization

As discussed in Section 3.3, one of MoBES's key strengths lies in its *interpretability*, i.e., the metric importance of Equation 3 allows for the decomposition of exposure scores into contributions from individual metrics. This enables both the measurement and ranking of how much each metric influences a user's exposure. While these rankings can vary significantly across users, reflecting the inherent heterogeneity in mobility behaviors, they provide valuable insight into user exposure and, consequently, into the design of targeted protection mechanisms, tailored to the specific metrics that most affect a given user or user group.

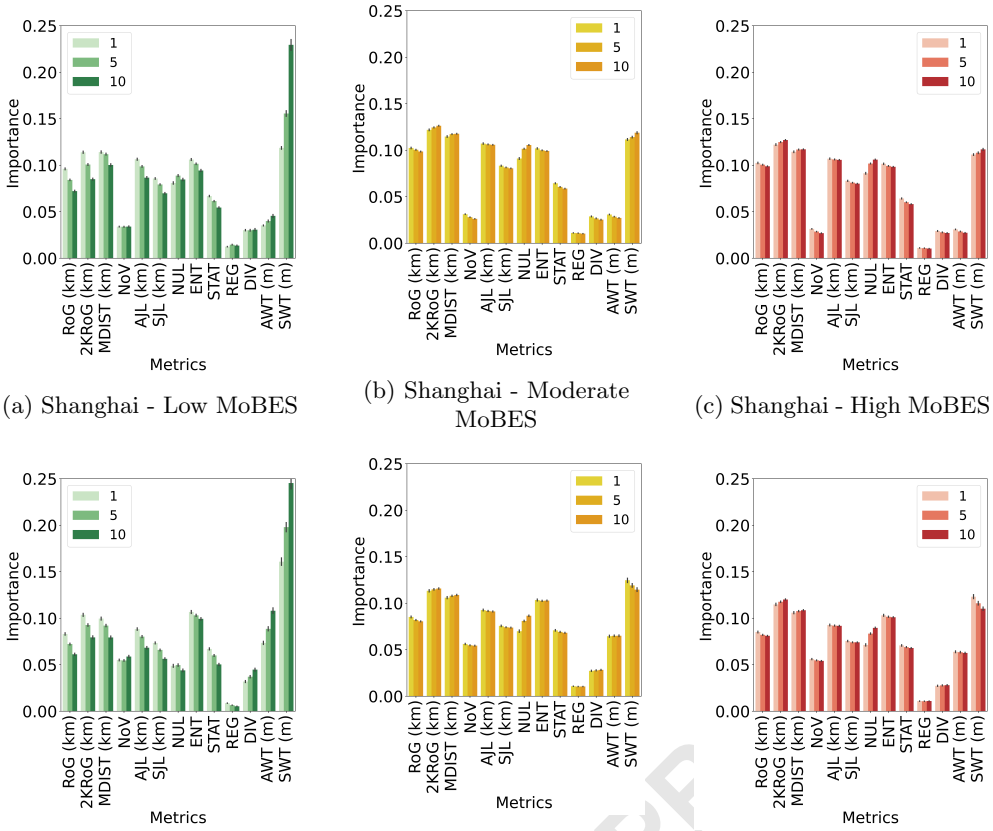
Indeed, the variability in metric importance across users is striking. In Shanghai, over 54,000 out of 58,502 users exhibit a unique ranking of metric contributions when $k = 10$; i.e., for over 92% of users, the order in which the mobility metrics contribute to their MoBES score is distinct from that of other users. Similarly, in Shenzhen, more than 29,200 out of 31,859 users have unique rankings. These results highlight the difficulty of manually interpreting metric influence at scale. Nevertheless, some general trends emerge: certain metrics tend to consistently rank high in importance, while others appear more frequently at the bottom.

Thus, to facilitate our analysis and discussion, we categorize users into three groups based on their MoBES values: **low MoBES** ($MoBES_i \leq 0.05$), **moderate MoBES** ($0.05 \leq MoBES_i < 0.10$), and **high MoBES** ($MoBES_i \geq 0.10$). Users in the high MoBES group are the more exposed ones. These thresholds were defined according to the MoBES distributions in our datasets and may need to be adjusted when analyzing other mobility datasets with different characteristics.

Figure 4 shows the average importance of each metric, according to Equation 3, along with 95% confidence intervals, for each user group, dataset, and value of k. Higher values indicate a greater contribution of the metric to users' overall exposure, as measured by MoBES. Figure 5 complements these results by reporting the proportion of users in each exposure category for both datasets. As expected from the distributions in Figure 3, most users fall into the low or moderate MoBES categories, in both datasets. However, we also observe that, proportionally, users in Shenzhen are more likely to fall into one of the extreme MoBES groups (either low or high). In contrast, users in Shanghai are more concentrated in the moderate MoBES range.

Our analysis in next sections is also informed by the results in Table 3, which reports the average values and 95% confidence intervals of all metrics across user groups and datasets. For clarity, we focus on results for k equal to 1 and 10, as the main trends are consistent for $k = 5$ as well.

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(a) Shanghai - Low MoBES

(b) Shanghai - Moderate MoBES

(c) Shanghai - High MoBES

(d) Shenzhen - Low MoBES

(e) Shenzhen - Moderate MoBES

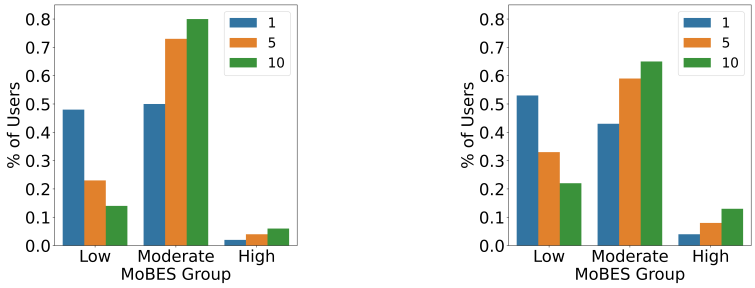
(f) Shenzhen - High MoBES

Fig. 4: Feature importance scores $I_{i,r}$ (Equation 3) for varying neighborhood sizes $k = \{1, 5, 10\}$, stratified by MoBES level (low, moderate, high) in Shanghai (a–c) and Shenzhen (d–f). We notice that some metrics, such as Regularity and Diversity, have a consistently low contribution to capturing exposure, while the importance of the other metrics varies according to the scenario.

5.2.1 General trends

Before delving into our group-level analysis, we first highlight some general trends observed in Figure 4 and Table 3, which apply across all three user groups.

First, as shown in Figure 4, the structural metrics Regularity (REG) and Diversity (DIV) consistently rank among the least important across all user groups, values of k, and datasets. This low importance stems from the fact that users tend to exhibit very similar behavior with respect to these metrics, regardless of their MoBES level, as indicated by the narrow confidence intervals in Table 3. Consequently, REG and DIV offer limited discriminative power within groups. This highlights that the behavioral space could potentially be reduced as some of the metrics have a low contribution.



(a) Shanghai (b) Shenzhen

Fig. 5: Proportion of users classified into different groups, namely low, moderate, and high MoBES, for k=$\{1, 5, 10\}$ in (a) Shanghai and (b) Shenzhen. In both datasets, most users fall into the low and moderate groups, but Shenzhen contains a larger proportion of users in the high MoBES group, reflecting greater behavioral heterogeneity.

Nonetheless, we also observe that these metrics allow a deeper interpretability of the users' mobility, being important for the discussions made in the next sections.

A similar pattern is observed for Number of Visits (NoV) and Average Waiting Times (AWT), which also appear among the least important metrics in both datasets. Their importance is particularly low in Shanghai, likely due to the (mostly) hourly sampling rate, which results in more uniform tracking of user movement and, hence, lower variability across users (reflected in Shanghai's tighter confidence intervals for both metrics, shown in Table 3). In contrast, the event-driven data collection in Shenzhen captures more variable user behavior on these metrics, giving NoV and AWT slightly higher relevance for quantifying user exposure (but still among the least important metrics for assessing user exposure).

Second, spatial metrics such as Radius of Gyration (RoG) and Maximum Displacement (MDIST) show a positive association with MoBES in both cities. This suggests that more exposed users tend to travel longer distances and have less confined mobility, i.e., are big travelers. In contrast, the 2-Radius of Gyration (2kRoG) shows no consistent trend, especially in Shanghai, likely because it focuses on regular home–work routines, which vary widely between individuals and offer less reliable signals of exposure. A similar pattern is observed for Number of Unique Locations (NUL) in Shanghai. Although users with higher MoBES in that city tend to be more travelers, they also tend to have fewer visits captured (see lower NoV in Table 3), which in turn results in higher average values for both temporal metrics, i.e., Average Waiting Times (AWT) and Standard deviation of Waiting Times (SWT). This contrasts with Shenzhen, where users with higher values of MoBES tend to have more visits captured (higher NoV), leading to lower AWT and SWT values.

Next, we proceed to a detailed examination of user behavior, focusing on the most important metrics for exposure in each user group, i.e., Low, Moderate, and High MoBES.

Table 3: Average values (and 95% confidence intervals (CIs)) for the metrics of users in different MoBES groups - low, moderate, high (95% CIs marked with ⋆ have margins of error below 0.01). Higher MoBES is generally associated with broader spatial mobility, observed especially in the RoG metric, and larger variability across users, observed in wider CIs.

MoBES group	Metric	Shanghai		Shenzhen	
		k = 1	k = 10	k = 1	k = 10
Low	RoG (km)	3.19±0.04	3.22±0.07	0.66±0.03	0.3±0.01
	2KRoG (km)	2.31±0.04	3.04±0.07	0.58±0.03	0.27±0.01
	MDIST (km)	13.96±0.15	9.57±0.21	1.88±0.06	0.85±0.03
	NoV	181.15±0.05	181.26±0.1	131.85±0.56	131.35±0.88
	AJL (km)	1.21±0.02	1.16±0.03	0.38±0.01	0.19±0.01
	SJL (km)	2.77±0.03	2.47±0.05	0.59±0.02	0.27±0.01
	NUL	11.62±0.08	9.89±0.15	7.37±0.09	4.89±0.1
	ENT	0.99±0.01	0.87±0.01	0.98±0.01	0.78±0.01
	STAT	0.74±0.00⋆	0.8±0.00⋆	0.7±0.00⋆	0.77±0.01
	REG	0.94±0.00⋆	0.95±0.00⋆	0.94±0.00⋆	0.96±0.00⋆
	DIV	0.83±0.00⋆	0.68±0.01	0.75±0.00⋆	0.57±0.01
	AWT (min)	79.57±0.02	79.52±0.04	166.81±0.56	167.47±0.89
	SWT (min)	37.34±0.05	36.97±0.1	256.41±1.24	258.06±1.96
Moderate	RoG (km)	3.79±0.05	3.46±0.03	1.65±0.04	1.13±0.03
	2KRoG (km)	1.51±0.04	1.72±0.03	1.04±0.04	0.9±0.03
	MDIST (km)	20.42±0.2	18.11±0.14	5.63±0.13	3.42±0.08
	NoV	179.99±0.09	180.66±0.06	140.95±0.64	135.56±0.51
	AJL (km)	1.43±0.02	1.32±0.01	0.89±0.02	0.64±0.02
	SJL (km)	3.55±0.04	3.21±0.03	1.56±0.04	1.02±0.03
	NUL	13.27±0.09	12.92±0.06	12.17±0.15	10.43±0.1
	ENT	1.07±0.01	1.07±0.00⋆	1.2±0.01	1.16±0.01
	STAT	0.71±0.00⋆	0.71±0.00⋆	0.63±0.00⋆	0.63±0.00⋆
	REG	0.93±0.00⋆	0.93±0.00⋆	0.91±0.00⋆	0.92±0.00⋆
	DIV	0.88±0.00⋆	0.89±0.00⋆	0.89±0.00⋆	0.88±0.00⋆
	AWT (min)	80.05±0.04	79.77±0.02	154.98±0.62	161.64±0.51
	SWT (min)	38.39±0.07	37.94±0.05	223.36±1.39	242.15±1.13
High	RoG (km)	5.35±0.39	5.12±0.18	3.28±0.2	2.86±0.1
	2KRoG (km)	1.67±0.24	1.67±0.12	0.99±0.12	1.13±0.07
	MDIST (km)	26.63±1.5	26.01±0.73	13.99±0.68	11.4±0.32
	NoV	169.98±1.21	174.9±0.52	151.55±2.36	149.23±1.22
	AJL (km)	1.96±0.19	1.88±0.09	1.46±0.1	1.37±0.05
	SJL (km)	4.71±0.32	4.65±0.15	3.29±0.18	2.85±0.09
	NUL	11.95±0.68	12.22±0.36	14.0±0.75	13.72±0.36
	ENT	1.02±0.05	0.97±0.02	1.18±0.04	1.19±0.02
	STAT	0.72±0.01	0.73±0.01	0.64±0.01	0.64±0.01
	REG	0.93±0.00⋆	0.93±0.00⋆	0.9±0.01	0.9±0.00⋆
	DIV	0.74±0.02	0.77±0.01	0.86±0.01	0.87±0.00⋆
	AWT (min)	84.41±0.53	82.27±0.23	142.86±2.09	146.37±1.11
	SWT (min)	43.23±0.93	40.44±0.36	189.87±4.57	198.73±2.44

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5.2.2 Low MoBES ($MoBES_i \leq 0.05$):

Users in this group display behavioral patterns that closely resemble those of their nearest neighbors: a deviation of just 5% is enough to blend them among the neighbors. Many of these users tend to be highly stationary, especially in Shenzhen, often

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staying in the same location for extended periods. This is reflected in higher values of stationarity (STAT) compared to users in the other two groups (see Table 3).

Among those who are more mobile, movement is typically constrained to home–work routines, with a few other occasional visits, particularly in Shanghai. This pattern results in a smaller difference between RoG and 2kRoG than in the other groups, as shown in Table 3. Likewise, the difference between RoG and MDIST tends to be narrower as well, especially in Shenzhen, indicating fewer (and shorter) long-distance transitions between consecutive locations.

Overall, for Shenzhen users, spatial metrics such as MDIST, RoG, as well as Average and Standard deviation of jump lengths (AJL and SJL) are not only lower in magnitude but also more similar across users, as shown by their smaller means and tighter confidence intervals in Table 3. The higher stationarity also leads to lower entropy (ENT).

Given this overall behavioral homogeneity, *variability in Waiting Times captured by the SWT metric emerges as the most important factor for distinguishing among users in this group, and thus for assessing behavioral exposure* (see Figure 4). This holds across both datasets and all values of k, with SWT becoming increasingly important as k increases. As more neighbors are included in the MoBES computation, the users who continue to exhibit low MoBES scores tend to be those with the lowest mobility and highest similarity in terms of spatial metrics. In such cases, temporal irregularities become even more crucial for exposure assessment, further amplifying the importance of SWT.

5.2.3 Moderate ($0.05 \leq MoBES_i < 0.1$)

Users in this group exhibit more traveler behavior compared to those in the Low MoBES group. As shown in Table 3, the Number of unique locations visited (NUL) increases modestly in Shanghai and substantially in Shenzhen, registering approximately a 65.15% to 113.29% increase for the two values of k. Similarly, other spatial metrics, including RoG, MDIST, AJL, and SJL also exhibit higher values, suggesting greater spatial coverage. Moreover, the structural metrics DIV and ENT also increase, reflecting greater mobility diversity.

Notably, the increased gap between RoG and 2kRoG suggests that users in this group tend to move well beyond their home and work locations, reinforcing their higher traveler nature of their mobility patterns.

This increased heterogeneity naturally results in higher exposure compared to the Low MoBES group. As illustrated in Figure 4, *SWT remains one of the most important metrics for assessing user exposure.* However, unlike in the Low MoBES group, *the importance of several spatial metrics rises (modestly) due to the broader traveler nature.* As a result, the relative importance of SWT drops, becoming comparable to that of the spatial metrics, excluding NoV (which, as discussed in Section 5.2.1, is among the least important metrics for all user groups).

5.2.4 High $MoBES_i \geq 0.1$

Users in this group exhibit higher traveler behavior than those in the Moderate and Low MoBES group. As shown in Table 3, they have higher values for several spatial

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metrics, particularly RoG, MDIST, AJL, and SJL, indicating broader spatial coverage. In addition, these metrics tend to have wider confidence intervals compared to those in the Moderate group, signaling greater variability across users. *This heterogeneity in mobility behavior contributes directly to their higher exposure*, as reflected by increased MoBES values. As shown in Figure 4, the relative importance of the various metrics remains similar to those of the Moderate MoBES groups, with user exposure being driven mainly by SWT and the spatial metrics.

In Shenzhen, specifically, users in the High MoBES group also exhibit higher values of NUL and NoV, along with lower values for both AWT and SWT. These patterns suggest that users not only visit more locations, but also explore new ones more frequently, resulting in shorter Waiting Times between movements. Again, the broader confidence intervals for these metrics highlight greater behavioral diversity, reinforcing the link between mobility heterogeneity and exposure.

In Shanghai, by contrast, users in this group display higher values of AWT and SWT. This is primarily due to data sparsity, discussed in Section 5.2.1, and is reflected in lower values of NoV, NUL, and DIV. Yet, even in this more limited setting, the variability across users increases, as indicated by the wider confidence intervals. *This continued rise in user heterogeneity, even under sparse data conditions, remains a key driver of higher exposure.*

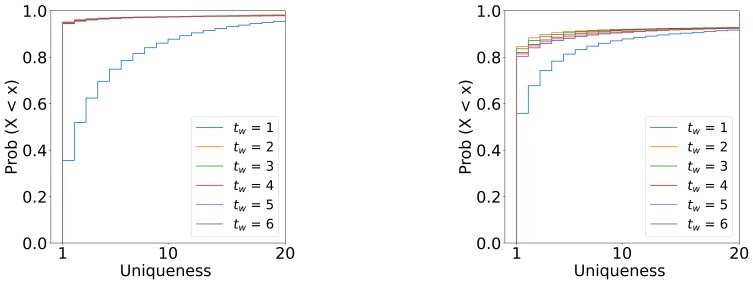
In summary, this section highlights the strong interpretability and observability of MoBES in capturing differences in user mobility behavior across varying levels of behavioral exposure. In the following section, we move beyond exposure driven by behavioral patterns and compare MoBES with a widely used alternative measure of exposure, namely Uniqueness.

6 Comparing exposure measures: *Uniqueness* vs. MoBES

In this section, we compare MoBES with the *Uniqueness* measure proposed by De Montjoye et al. [6]. As discussed in Section 2.2, *Uniqueness* quantifies user exposure based on the unicity of the users' spatial-temporal patterns within a dataset. Specifically, it considers all ordered subsequences of spatiotemporal points within a time window of duration t_w. A user is assigned a *Uniqueness* value of 1 if at least one such subsequence is unique, indicating full exposure during that time window.

While both MoBES and *Uniqueness* aim to assess user exposure, they capture distinct perspectives of it. MoBES measures the unicity in the users' behavior using a combination of mobility metrics. *Uniqueness*, in contrast, focuses on the unicity of specific spatiotemporal sequences within fixed time intervals. Thus, our goal in this section is to examine the overlap between users identified as exposed by both measures, as well as those flagged by only one. This analysis sheds light on how each measure captures different aspects of exposure in mobility data.

Importantly, while both MoBES and *Uniqueness* produce continuous-valued exposure scores, prior studies using *Uniqueness* have predominantly applied it in a binary



(a) Shanghai (b) Shenzhen

Fig. 6: Cumulative distributions of *Uniqueness* under varying temporal aggregation windows $t_w = \{1, 2, 3, 4, 5, 6\}$ for (a) Shanghai and (b) Shenzhen. Except for $t_w = 1$, *Uniqueness* identifies a very large fraction of users as exposed, particularly in Shanghai, due to the larger area of observability and frequency in data collection.

classification setting [7]. Following this convention, we also perform a comparison considering two levels of exposure: exposed vs. not exposed. We define the set $Uniq^*$ to include all users with a *Uniqueness* score of 1 (i.e., exposed according to *Uniqueness*), mirroring the notation of $MoBES^*$ (defined in Section 3.1), which includes users with $MoBES_i > \tau$ (i.e., exposed users according to MoBES);

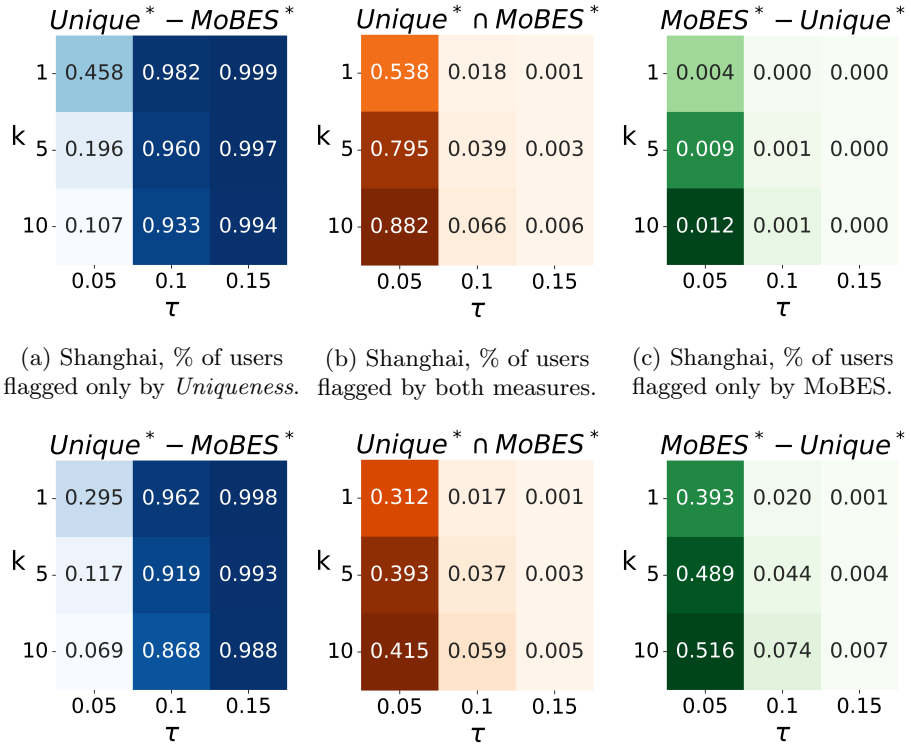
We begin by presenting the *Uniqueness* results for both the Shanghai and Shenzhen datasets (Section 6.1), followed by a direct comparison between the two measures (Section 6.2).

6.1 *Uniqueness* in mobility

Figure 6 presents the distributions of *Uniqueness* values across different aggregation time windows t_w for both Shanghai and Shenzhen datasets. The figure reveals a high level of user exposure in both cities: the fraction of exposed users in $Uniq^*$, reaching nearly 100% in Shanghai and approximately 80% in Shenzhen. These findings confirm that subsequences of visited locations within particular time windows are indeed strong indicators of user exposure in mobility datasets.

The observed differences in exposure, as captured by *Uniqueness*, between the two datasets reflect both geographical and user behavioral factors. The Shanghai dataset covers a significantly larger area of observation (see Figure 1), which increases the likelihood of users generating unique movement patterns, especially over longer distances. This broader spatial scope helps explain the exposure of almost the entire population. User behavior also contributes to the difference. As previously discussed, users in Shanghai primarily moving between home and work locations, tend to exhibit greater displacements, evidenced by higher RoG, MDIST, AJL and SJL values, compared to users in Shenzhen (see Figure 2). This broader range of mobility further increases the likelihood of unique subsequences in the trajectory, and hence, higher exposure.

For the remainder of the analysis, we adopt a fixed window size of $t_w = 4$, since increasing t_w yields diminishing returns in terms of additional user exposure.



(a) Shanghai, % of users flagged only by *Uniqueness*.

(b) Shanghai, % of users flagged by both measures.

(c) Shanghai, % of users flagged only by MoBES.

(d) Shenzhen, % of users flagged only by *Uniqueness*.

(e) Shenzhen, % of users flagged by both measures.

(f) Shenzhen, % of users flagged only by MoBES.

Fig. 7: Fractions of users flagged as exposed by MoBES and/or *Uniqueness*, for various MoBES parameter settings (parameters k and τ). Fractions computed over the total number of exposed users according to either measure ($Uniq^* \cup MoBES^*$). The substantial fraction of users identified by only one of the two measures demonstrates that MoBES and Uniqueness capture complementary dimensions of exposure rather than redundant information.

6.2 Benchmarking MoBES against *Uniqueness*

We now turn to a comparison between MoBES and Uniqueness. Before presenting the results, we stress that both measures provide estimates of user exposure grounded in distinct methodological assumptions, and no ground truth is available for validation. Consequently, any claims regarding differences in the set of exposed users captured by each measure must be interpreted within the scope of these underlying assumptions.

To compare MoBES and *Uniqueness*, we look into users identified as exposed by either or both measures. To ensure a comprehensive comparison, we evaluate MoBES under different parameter settings, using values of $k = 1, 5, 10$ and $\tau = 0.05, 0.1, 0.15$. We organize our analysis around three mutually exclusive subsets of users:

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Table 4: Numbers of users flagged as exposed only by *Uniqueness*, by both measures, and only by MoBES, for various MoBES parameter settings (k and τ). The results reveal an intersection of users captured by both *Uniqueness* and MoBES, being especially high when $\tau = 0.05$. Moreover, it also reveals the number of users that are captured only by one of the measures, *Uniqueness* on the left, and MoBES on the right, showing their complementarity.

		$Unique^* - MoBES^*$			$Unique^* \bigcap MoBES^*$			$MoBES^* - Unique^*$		
City	k/τ	0.05	0.10	0.15	0.05	0.10	0.15	0.05	0.10	0.15
Shanghai	1	25,576	54,620	55,551	30,037	993	62	249	11	1
	5	11,008	53,434	55,437	44,605	2,179	176	489	34	1
	10	6,003	51,946	55,303	49,610	3,667	310	654	67	3
Shenzhen	1	12,683	25,638	26,075	13,417	462	25	16,869	542	38
	5	6,001	25,101	26,024	20,099	999	76	24,995	1,214	101
	10	3,712	24,451	25,959	22,388	1,649	141	27,876	2,085	172

1. $Uniq^* - MoBES^*$: users flagged as exposed only by *Uniqueness*;
2. $Uniq^* \cap MoBES^*$: users flagged as exposed by both *Uniqueness* and MoBES;
3. $MoBES^* - Uniq^*$: users flagged as exposed only by MoBES.

Figure 7 presents the distributions of users across the three subsets for both datasets and across the various parameter combinations. It is important to note that these proportions are calculated for the set of all users considered exposed by either measures, i.e., users in $Uniq^* \cup MoBES^*$, rather than the total population. Users not flagged as exposed by either method are excluded from this calculation. As an example, in the Shanghai dataset with $k = 1$ and $\tau = 0.05$, approximately 45.8% of exposed users in $Uniq^* \cup MoBES^*$ are flagged as exposed only by *Uniqueness*, both methods flag 53.8%, and only 0.4% are flagged solely by MoBES. Complementarily, Table 4 presents the absolute values of the percentage captured in the Figure 7.

Before examining the specific characteristics of each subset, we first highlight some key observations that are central to interpreting the comparison. We then analyze the groups of users flagged as exposed exclusively by one of the measures, as well as by both. For completeness, we also briefly discuss users considered protected (i.e., not exposed) by both measures. This analysis builds on the data and mobility patterns discussed in Sections 4 and 5.

6.2.1 Key observations

Based on the results in Figure 7, we make four key observations:

First, several parameter configurations, a meaningful fraction of users are flagged as exposed by only one of the two measures (Figures 7a, 7d, 7f). This strongly suggests that MoBES and *Uniqueness* are complementary. In particular, this leads to two important conclusions: (i) each method seems to capture a different perspective of user exposure, and (ii) these perspectives reinforce each other when used jointly to assess user exposure in mobility data. Notably, in scenarios with a more flexible

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notion of exposure, such as smaller values of k and larger values of τ, a significant number of users are flagged as exposed only by MoBES, especially in Shenzhen. In contrast, this effect is less prominent in Shanghai, where users tend to exhibit more heterogeneous mobility patterns, making them more easily detectable as exposed by both measures. These findings highlight a frequently overlooked aspect of exposure: user mobility behavior, which goes beyond merely identifying where and when users move. MoBES aims at capturing this behavioral dimension of exposure: one that may remain hidden when relying solely on the raw location sequences used by *Uniqueness*.

Second, for a fixed value of k, increasing τ makes MoBES adopt a more flexible notion of exposure, thus becoming more selective, that is, users must exhibit more distinctive behavior to be flagged as exposed. As a result, the number of users in $MoBES^*$ decreases (and so does the number of users flagged by both measures), while the fraction of users flagged only by *Uniqueness* increases.

Third, for a fixed τ, increasing k has the opposite effect. A larger k dilutes individual behavior by blending it with that of more neighbors, which in turn inflates the number of users flagged by MoBES (either in isolation or jointly with *Uniqueness*). For example, with $k = 10$ and $\tau = 0.05$, most users flagged by *Uniqueness* are also flagged by MoBES (compare Figures 7b and 7a, as well as Figures 7e and 7d. In Shenzhen, this configuration results in 51% of users being flagged as exposed only by MoBES (see Figure 7f). In Shanghai, in turn, only 1% of users are flagged as exposed only by MoBES in the same configuration since, as discussed, the greater heterogeneity of mobility patterns leads to practically all users being flagged as exposed by *Uniqueness*.

The two previous observations illustrate how parameter choices critically shape MoBES's notion of exposure and, consequently, its divergence from *Uniqueness*: *increasing k and decreasing τ (i.e., reflecting a stricter or more sensitive exposure policy) broaden MoBES's scope, allowing it to identify a larger set of exposed users not captured by Uniqueness.*

Finally, **fourth**, city-specific mobility patterns are key to MoBES's ability to capture exposure patterns, especially among users not identified as such by *Uniqueness*. In Shenzhen, where users tend to move within a smaller spatial area, MoBES uncovers a larger share of exposed users. The limited spatial coverage reduces the chance of generating unique spatiotemporal subsequences, making *Uniqueness* less effective in that context. In contrast, the broader observation area and greater user mobility characteristic of Shanghai increases the chance of users having unique subsequences of visited locations, thus being flagged as exposed by *Uniqueness*.

Keys summary: These findings underscore that *Uniqueness should not be treated as a baseline for MoBES, but rather as a complementary perspective on user exposure.* While *Uniqueness* focuses on the distinctiveness of raw spatiotemporal trajectories, MoBES offers a behavioral lens, analyzing patterns derived from those trajectories. *Together, they provide a more comprehensive understanding of exposure in mobility datasets.*

6.2.2 Users flagged as exposed only by *Uniqueness*

In this section, we analyze users flagged as exposed only by *Uniqueness*, i.e., users in $Uniq^* - MoBES^*$. These users exhibit low MoBES scores (i.e., are behaviorally

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protected) but low *Uniqueness* scores (i.e., are spatially exposed). In other words, despite following unique displacement patterns, their overall mobility behavior, captured through MoBES, is typical among peers, explaining why they are not flagged as exposed by MoBES.

As discussed in Section 5, a common trait among these users is that most of their displacements occur around home and work. The key difference across datasets lies in users travelled distances: in Shanghai, for instance, users tend to travel longer distances often. As a result, even if they visit the same locations, they may have distinct sub-trajectories during specific time periods, leading them to be exposed according to *Uniqueness*. However, because their overall behavioral profile remains similar (e.g., high stationarity), they stay unflagged by MoBES.

To further illustrate such differences, Figures 8a and 8b present an illustrative example from the Shenzhen dataset. We consider three users: (i) u_i (target user), flagged as exposed only by *Uniqueness* (in red); (ii) u_i's spatial neighbor, i.e., the neighbor whose trajectory most closely overlaps with u_i's, potentially offering spatial protection, but ultimately not doing so (in orange); and (iii) u_i's behavioral neighbor, i.e., u_i's nearest neighbor in MoBES m-dimensional behavioral space (in green).

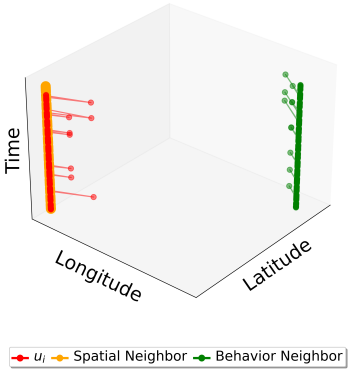
In Figure 8a, we present a 3D scatter plot depicting the trajectories of the three users. The x and y axes represent spatial coordinates, while the z axis captures the temporal dimension: higher values of z indicate later points in time. Both the target user u_i (in red) and her spatial neighbor (in orange) exhibit high stationarity, remaining in the same location (both users) most of the time. This is illustrated by the dense orange vertical line, which also overlaps with the red points, indicating that both users spend most of their time in the same place. However, u_i occasionally deviates from this routine by visiting nearby locations (as shown by the red points scattered to the left of the orange line). These deviations, although minor, are not mirrored by the spatial neighbor and lead to unique spatial subsequences, causing u_i to be flagged as exposed according to the *Uniqueness* measure.

In contrast, the behavioral neighbor (in green) shows a similar overall mobility pattern to u_i: predominantly stationary with occasional visits to a secondary location. This similarity in behavioral dynamics becomes evident in Figure 8b, which shows a radar chart comparing the behavioral metrics of the three users (normalized to [0,1] for visualization). The target user u_i and her behavioral neighbor have great similarity across all metrics, highlighting their similarity in mobility behavior and explaining why MoBES does not flag u_i.

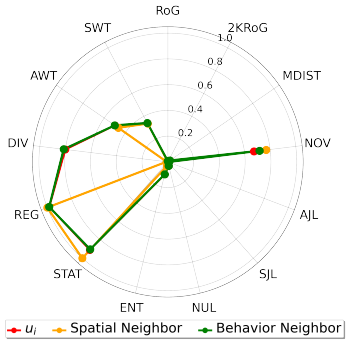
6.2.3 Users flagged as exposed only by MoBES

We now analyze users flagged as exposed only by MoBES, i.e., users in $MoBES^* - Uniq^*$. These users exhibit behavioral patterns that are markedly different from their peers, resulting in high MoBES scores. As discussed in Section 5, users with higher values of MoBES tend to be big travelers with wider spatial mobility, deviating from home and work by making larger displacements and visiting new locations.

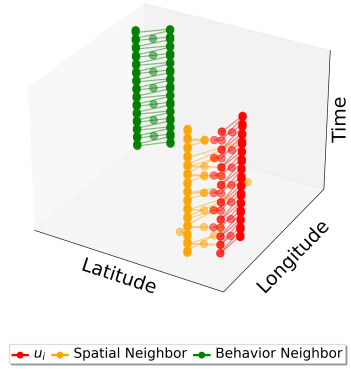
Despite the higher heterogeneous mobility patterns, these users are not flagged by the *Uniqueness* measure because their trajectories are not unique: they often visit popular or high-traffic areas (e.g., malls) where they are not alone in space-time. A



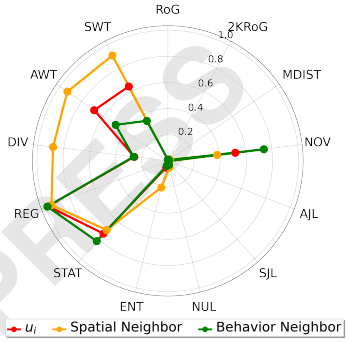
(a) Spatial-temporal movements of user $u_i \in Unique^*$ - $MoBES^*$.



(b) Mobility behavioral patterns of user $u_i \in Uniq^* - MoBES^*$.



(c) Spatial-temporal movements of user $u_i \in MoBES^*$ - $Uniq^*$.



(d) Mobility behavioral patterns of user $u_i \in MoBES^*$ - $Uniq^*$.

Fig. 8: Illustrative examples of selected users in the Shenzhen dataset. Each figure shows a user u_i, target of the analysis (in red), and their closest spatial (in orange) and behavioral (in green) neighbors. Figures (a) and (b) show a user u_i who is flagged as exposed only by Uniqueness: u_i exhibits spatio-temporal visitation patterns that, despite similarities, remain distinguishable from those of the closest spatial neighbor (a) , resulting in exposure according to Uniqueness; the same user shares highly similar behavioral characteristics with their closest behavior neighbor (b), leading to a low MoBES score. Figures (c) and (d) present another user u_i flagged only by MoBES: u_i has substantial overlap in terms of spatio-temporal visitations with nearby user (c), preventing exposure according to Uniqueness, but their behavioral profile differs markedly from that of the closest behavior neighbor (d), resulting in high behavioral exposure. Overall, the figures illustrates, once again, the complementarity of MoBES and Uniqueness. Note that the latitude and longitude axes of Figures (a) and (c) have been switched for the sake of improving readability.

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useful analogy is that of tourists in a crowded city: although their behavior (exploration, movement patterns, timing) may differ greatly from the local population, their physical presence in well-visited places at common times means they are not spatially unique, and thus, are protected under *Uniqueness*.

To better illustrate this case, we present an example of a Shenzhen user, shown in Figures 8c and 8d. The selected user u_i (in red) has a discernible MoBES score ($MoBES_{u_i} = 0.20$) and a *Uniqueness* score higher than 1. As shown in the spatial-temporal plot (Figure 8c), u_i and her spatial neighbor (in orange) follow broadly similar routines, frequently returning to home and work locations, with occasional exploratory visits. Although the spatial neighbor does not precisely mirror u_i's trajectory, the overlap in their movements is sufficient to shield u_i from exposure under the *Uniqueness* criterion.

In contrast, the behavioral patterns tell a different story. As shown in Figure 8d, u_i diverges significantly from both her spatial and behavioral neighbors. Of particular interest is the divergence between u_i and her behavioral neighbor, which is more pronounced, especially along the temporal metrics. These differences place u_i at a considerable distance from her neighbors in the m-dimensional behavioral space, thus rendering u_i exposed according to MoBES.

6.2.4 Users flagged similarly by both *Uniqueness* and MoBES

Finally, we examine the subsets of users who were flagged similarly, i.e., either exposed (i.e., in $Uniq^* \cap MoBES^*$ set) or not, by both measures.

Users flagged as exposed by both measures are those who display wider spatial mobility behavior, frequently visiting new locations. As discussed in Section 5, their behavioral patterns are more heterogeneous, setting them apart from their peers and leading to high exposure under MoBES. This behavioral diversity, combined with wider spatial mobility, also results in more unique spatio-temporal sub-trajectories, making them exposed according to *Uniqueness* as well.

Conversely, users not flagged as exposed by either measure, i.e., protected users under both MoBES and *Uniqueness*, tend to exhibit low mobility and high stationarity around routine locations such as home and work. These locations, in turn, tend to be densely populated areas (e.g., large commercial centers), which increases the likelihood of their trajectories overlapping with those of many others. As a result, these users blend into the crowd and remain protected according to both measures. These users are mostly characterized by the low MoBES group in Table 3.

Key insights:

Our results suggest that MoBES complements the state-of-the-art *Uniqueness* measure by capturing dimensions of user exposure that *Uniqueness* may miss. While several users, especially in Shenzhen, appear protected under *Uniqueness*, sharing similar spatio-temporal patterns with others, MoBES reveals that their behavioral signatures can still be highly distinctive. This distinction highlights the potential for identifiability based on behavioral cues, even when spatial-temporal overlap suggests otherwise.

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7 Investigating the impact of defense mechanisms

In this section, we evaluate the impact of defense mechanisms on MoBES. **More specifically, we analyze how Differential Privacy (DP) techniques [21] designed for spatial data affect users' behavioral patterns.** We adopt DP due to its strong mathematical guarantees, which provide formal privacy protection [17]. Next, we briefly outline the mathematical framework of DP and describe the specific instantiation employed in this study. We then examine how the application of DP influences the underlying behavioral characteristics of users and consequently MoBES scores.

7.1 Differential Privacy

DP is a mathematical framework that quantifies and bounds privacy loss [40]. It enables the extraction, sharing, and analysis of patterns from datasets while providing formal guarantees on the privacy risks associated with individuals' participation. The core intuition of DP is that the inclusion or exclusion of a single individual's data (e.g., a row in a dataset) should not have a significant impact on the outcome of any analysis [17, 41]. As a result, an observer cannot reliably determine whether a specific individual's data was used in the computation. [21, 41]. Formally, DP is defined as follows: if any two datasets D_1 and D_2 differ by at most one record, then for all possible outputs S of a randomized mechanism κ, the following holds:

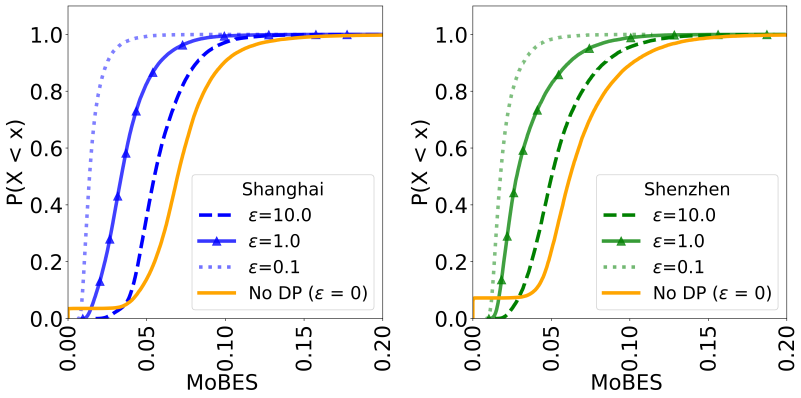
$$\mathbb{P}[\kappa(D_1) \in S] \leq e^{\varepsilon} \times \mathbb{P}[\kappa(D_2) \in S] \tag{4}$$

where the *privacy parameter* ε controls the strength of the guarantee, with *smaller values of* ε *providing stronger privacy and noisier output datasets that are harder to trace back to any specific user.* Typical values of ε range from 0.01 to 10 [42].

To ensure differential privacy, we adopt the C-Noise instantiation of the DP framework [21]. C-Noise perturbs each spatial coordinate of a trajectory by adding Laplace noise [43], treating coordinates as independently randomized values. Rather than releasing exact locations, the mechanism publishes perturbed coordinates, with the noise magnitude calibrated according to the privacy parameter ε, as defined in [21]. We refer the reader to the original work for a complete formal description of the mechanism and its privacy guarantees.

As discussed by Jiang et al. [21], the application of C-Noise to trajectory data involves a trade-off between privacy and trajectory utility, as the mechanism may introduce distorted patterns on the users trajectory. Nevertheless, we adopt C-Noise because it demonstrated better computational performance in the evaluation conducted by [17]. In our scenario, this trade-off is acceptable, since our primary focus is on assessing the impact of differential privacy on behavioral representations.

Lastly, we focus only on spatial protection, since adding temporal noise would greatly increase computational costs. Consequently, by choosing only spatial protection, the two temporal metrics (AWT and SWT) extracted from trajectories remain unchanged, as there is no change in the temporal dimension. Moreover, the metric



(a) Shanghai with Differential Privacy

(b) Shenzhen with Differential Privacy

Fig. 9: MoBES distributions for different values of ε and for k = 10. Stronger privacy protection (smaller ε) systematically reduces MoBES values, indicating that trajectory perturbation decreases behavioral exposure by making users more similar in the behavioral space. However, behavioral exposure is not completely eliminated.

NoV, which accounts for the number of records of a user, remains unchanged since we neither remove nor add new spatial-temporal points.

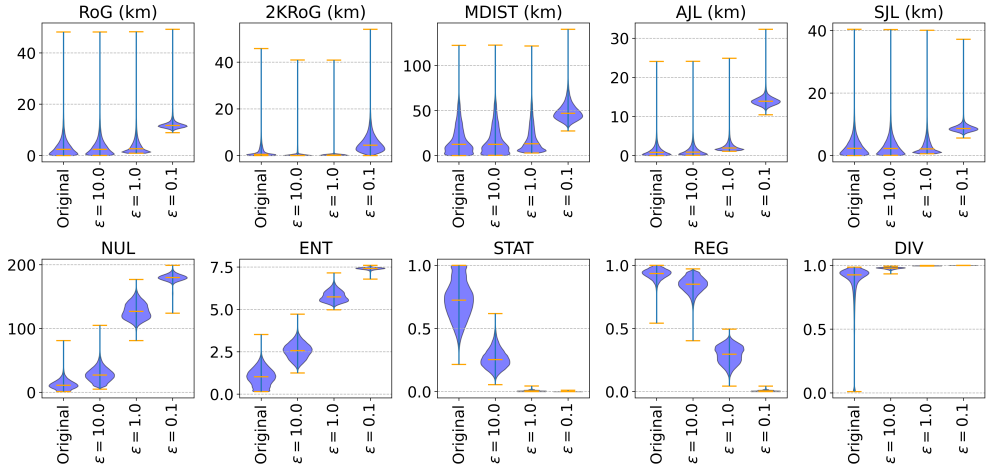
7.2 Impact of DP on MoBES

We now direct our focus to the effect of DP on MoBES. It is important to establish that we use as the parameter for the privacy budget (ε) values that are commonly applied in the literature, e.g., $\varepsilon = \{10, 1, 0.1\}$ [42]. Figure 9 illustrates, for $k = 10$, the distribution of MoBES values under varying levels of privacy protection[1]. As anticipated, MoBES decreases as the strength of the privacy mechanism increases (i.e., under small ε values). This reduction arises from the perturbation introduced by the protection mechanism, which affects the underlying metrics that constitute the behavioral space.

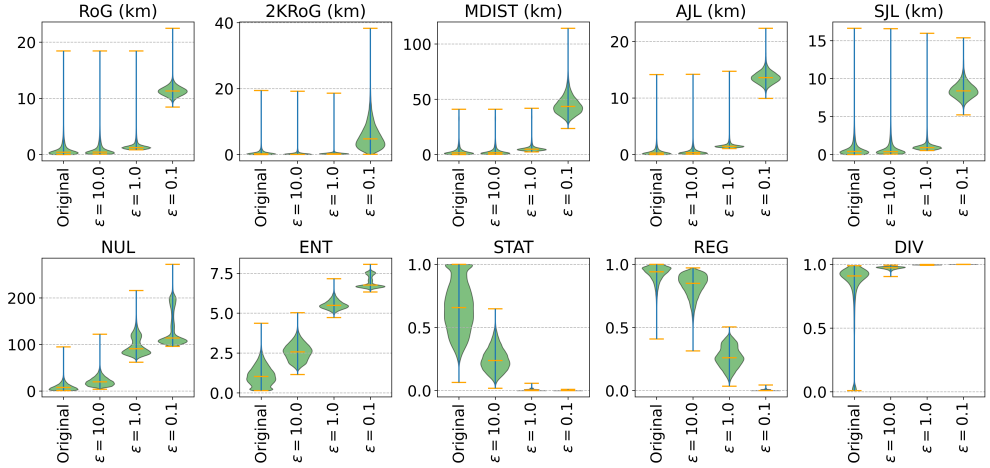
In Figure 10, which illustrates the impact of DP on the metrics, we observe that as higher levels of noise are applied (i.e., for smaller ε values), the metric distributions tend to concentrate around central values, thereby reducing variability. This effect is particularly pronounced for the Number of Unique Locations (NUL) and for the structural metrics Entropy (ENT), Stationarity (STAT), Regularity (REG), and Diversity (DIV). The reduction in variability causes users to become closer to one another in the m-dimensional behavioral space, thus decreasing their distance to nearest neighbors and impacting MoBES.

Additionally, NUL tends to increase as noise perturbs trajectory points. Since each pair of coordinates receives a different noise perturbation, an original point $l \in Tr_i$

[1] Only presented for $k = 10$ due to space constraints

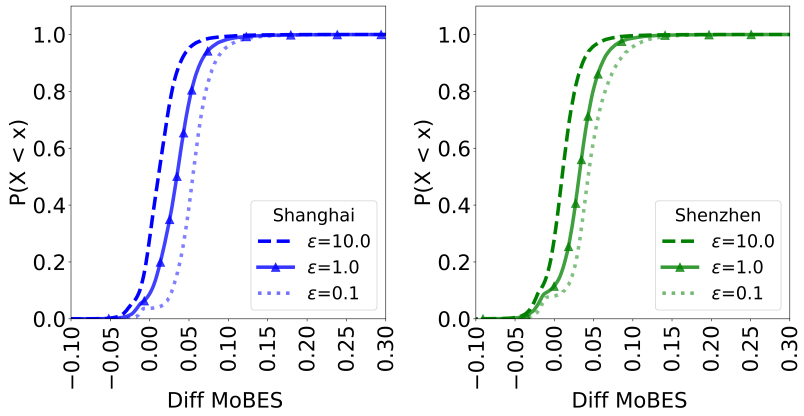


(a) Shanghai



(b) Shenzhen

Fig. 10: Distributions of the behavioral-space metrics for different levels of Differential Privacy (DP) protection. The figure illustrates the heterogeneous impact of DP on the considered metrics: the distributions remain close to the original data for most metrics, exhibiting substantial deviations only under strong privacy protection ($\varepsilon = 0.1$). In contrast, structural metrics and Number of Visits (NoV) exhibit pronounced distributional distortions even for less restrictive privacy settings. This behavior can be explained by the independent perturbation applied to each spatial coordinate, which may artificially displace users from their original locations and generate movement patterns even when users are initially stationary.



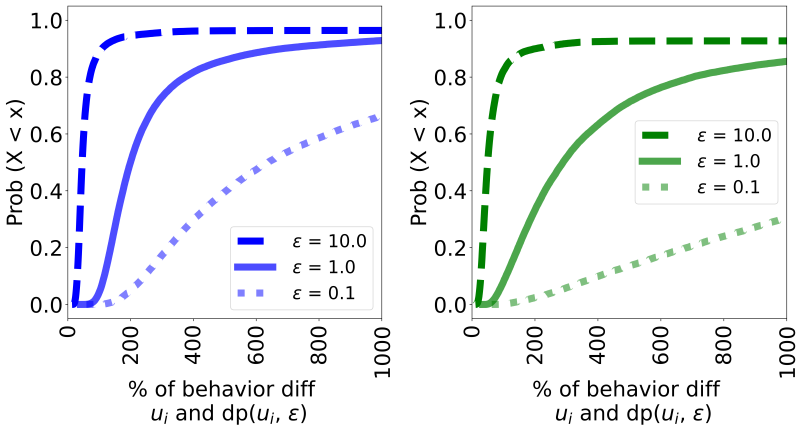
(a) Shanghai (b) Shenzhen

Fig. 11: Distributions of the differences between the original MoBES value and the MoBES value with the Differential Privacy, i.e., $MoBES_i - MoBES_i^{DP}$, for various ε and k = 10. Most differences are positive, particularly for stronger privacy settings, indicating that DP generally reduces behavioral exposure rather than creating new behavioral outliers.

may be transformed into multiple distinct visits. As a consequence, users' stationarity (STAT) and regularity (REG) decrease, while diversity (DIV) and entropy (ENT) increase.

Corroborating with our first claim that MoBES decreases as more protection is applied, we show in Figure 11 the distribution of the differences in MoBES values between the original MoBEs and MoBES with the DP, i.e., $MoBES_i - MoBES_i^{DP}$ We evaluate such distribution focusing on three main scenarios: (i) MoBES differences smaller than 0, meaning that the MoBES values increased, i.e., the use of DP revealed unique patterns of the user, thus increasing exposure; (ii) MoBES differences equal to zero, meaning that there was not clear impact from the use of DP; and (iii) MoBES differences bigger than 0, meaning that the MoBES values decreased, i.e., the user is more protected.

Figure 11 indicates that, in both datasets, only a small portion of the users present differences in MoBES values smaller than zero, while most of the users have a difference in MoBES values between 0 and 0.1. Such observation might suggest small differences in the overall behavior of the users, as reflected in the mobility metrics used, after DP is applied. However, this is not the case. These small differences emerge because the defense mechanism affects not only the original user but also most (if not all) the neighboring users, as the same noise is applied to everyone. This is illustrated in Figure 12, which shows the distribution of differences (in the behavioral space) between a user u_i and the same user after applying DP. Because DP is applied uniformly to all users with the same noise level ε, its impact on MoBES values depends directly on this parameter. As shown, in most cases, particularly for ε equal to 1 and 0.1, the differences are large (often above 100%). This suggests that DP effectively masks the



(a) Shanghai (b) Shenzhen

Fig. 12: Distributions of the relative difference in the behavioral space between a user before and after DP is applied, for various values of ε. Notice that by increasing ε, we also have an increase in the difference. Stronger privacy protection produces larger deviations in user behavior representations, demonstrating that the reduction in exposure comes at the cost of increasing distortion in the behavioral space.

original behavior of most users, as the behavioral metric values computed after its application differ substantially from the original ones.

In contrast, when a smaller noise level is used (e.g., $\varepsilon = 10$), the resulting changes in MoBES values are much smaller. Consequently, users who were exposed prior to the application of DP may remain exposed, as their behavioral metrics are only minimally altered. Overall, as the noise level increases, MoBES values tend to decrease, suggesting lower user exposure and, therefore, greater protection.

In summary, this section assessed how Differential Privacy affects MoBES by evaluating the impact on the individual metrics that underlie it. As key takeaways, we showed that MoBES values tend to reduce as more protection is applied to the data. This is directly linked to how DP impacts the metrics. Although some metrics are not affected, most of the metrics have a clear reduction in their variability, which makes the users closer to each other in those dimensions of the behavioral space. This is especially seen in structural metrics whose distributions differ markedly from their original ones, affecting the captured behavior. An encouraging aspect of the observed MoBES value reduction is that it suggests MoBES potential to capture user exposure accurately. As stronger protection mechanisms are applied, the exposure scores consistently decrease, reflecting the intended privacy gains. We contend that this also suggests MoBES could serve as a guiding metric for determining the optimal balance between utility and protection in mobility datasets, as it is derived from metrics whose utility diminishes as stronger privacy protection is applied. Importantly, user behavior is indeed increasingly protected as more noise is introduced into the data. However,

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when the applied noise is small, the resulting changes in behavioral metrics may be insufficient to effectively protect the user.

8 Design considerations in MoBES

In this section, we discuss important design considerations. While the measure provides a meaningful estimate of user exposure by capturing behavioral deviations from neighboring users, its effectiveness depends on key modeling choices, such as the aggregation of behavioral metrics and the distance function used to quantify dissimilarity. These design decisions influence both the sensitivity of MoBES to certain behavioral patterns and the interpretability of the resulting exposure scores. In the following, we discuss these aspects and explain the rationale behind the adopted choices.

Relative distance. One fundamental aspect of MoBES design is the distance metric. The choice of distance metric plays a central role in shaping the notion of behavioral dissimilarity captured by MoBES. While different metrics could be used, we adopt the *relative distance* because it provides a clear and interpretable meaning: it reflects how much a user would need to change their behavior, on average, to become closer to their nearest neighbors. This interpretation *facilitates the definition of exposure thresholds, supports more nuanced assessments of exposure levels, and allows direct comparison with Uniqueness.* Although the relative distance can accentuate differences in certain situations, this potential bias toward larger exposure values is desirable, as it leads to a more conservative assessment of exposure.

Averaging distances. Although MoBES effectively identifies users whose behavior deviates from that of their neighbors and provides a meaningful estimate of exposure, certain scenarios may reduce its sensitivity. Specifically, when the distance between a user and their neighbors is dominated by a single behavioral metric, the averaging used to aggregate distances across metrics may dilute this signal, potentially exposing the user to an attack exploiting that dimension. In such a scenario, MoBES may fail to properly flag the user's exposure. Methods tackling adversarial strategies based on attributes outside MoBES's behavioral representation are left for future work.

Interpretable τ parameter. While the relative distance may accentuate differences in certain situations, we favor it due to its interpretability, which is difficult to achieve with alternative distance metrics such as Euclidean, Manhattan, or Cosine. As previously discussed, this potential bias toward larger exposure values is desirable because it encourages a more conservative assessment of exposure. Moreover, a fair comparison between Uniqueness and MoBES configured with these alternative metrics would require redefining the threshold parameter τ, which must remain consistent with the chosen distance metric. In such cases, the lack of a clear interpretation of the MoBES score, resulting from the use of non-relative distances, would make selecting τ even more challenging. In other words, the *relative distance is a key component of our definition of exposure*, and although MoBES can be adapted to other metrics, doing so would require redefining the entire evaluation scenario.

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9 Conclusions and future work

This paper extends the evaluation of MoBES (Mobility Behavior-based user Exposure Score), a measure designed to assess and interpret user exposure in human mobility datasets. MoBES quantifies exposure by analyzing each user's mobility behavior relative to that of their nearest neighbors, identifying deviations that indicate higher exposure. By evaluating the users' behavior, which is modeled using different mobility metrics, MoBES is able to capture different perspectives from the measures available in the literature. Indeed, MoBES joins several characteristics of the works previously available in the literature as:

- **High-interpretability:** Allows for identifying the features that most impacted the user exposure, which can be further used by privacy protection methods as input for defining the level according to the user characteristics.
- **Continuous usability:** Allows defining for each user an exposure score, positioning the user in comparison to its peers.
- **Low computational cost:** Which is tied to the cost of measuring the k-nearest neighbors, being more scalable than measuring combinations of users' trajectory visits.
- **Easily customizable:** Allows easy customization of the behavioral space, specifically when side information is available.

In sum, MoBES is a tool that allows practitioners to evaluate the extent of exposure in their mobility datasets in a scalable way. Moreover, it opens the possibility for those who use the tool to address the users' exposure according to their level of exposure, potentially maximizing the data utility.

We dive into the evaluation of MoBES, assessing it with two CDR datasets namely Shanghai and Shenzhen, which were selected for their different spatial-temporal characteristics. While Shanghai is characterized by having a larger area of observation, with a more dense collection at every hour, Shenzhen has a smaller area with an event-based collection. Nevertheless, similar exposure patterns are observed in both Shanghai and Shenzhen: users classified as big travelers are typically more exposed. This is explained by the fact that, in both datasets, the predominant behavioral pattern consists of users whose mobility is largely confined to movements between home and work, with infrequent engagement in other activities.

Moreover, we position MoBES within the literature by comparing it with *Uniqueness*. The results show that MoBES is able to identify users who are also flagged by *Uniqueness*, suggesting that behavioral characteristics constitute a relevant indicator of exposure. Conversely, *Uniqueness* is also able to identify users who are not flagged by MoBES, indicating that it should not be viewed merely as a baseline. Instead, *Uniqueness* captures a complementary perspective to that of MoBES. Together, the measures enable the analysis of user exposure from both spatial-temporal and behavioral perspectives. Finally, our results show that MoBES is able to identify users who are behaviorally exposed but not spatially exposed, capturing a dimension of exposure that, within the scope of our experiments, is not detected by *Uniqueness*. This

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finding highlights a potential privacy threat that would otherwise remain unnoticed when relying solely on spatial-temporal exposure measures.

Lastly, we evaluated the impacts of Differential Privacy (DP) in the metrics that compose MoBES, i.e. the behavioral space, and in MoBES itself. Our results suggest that DP can protect users' against spatial-temporal exposure and behavior exposure. Nevertheless, the behavior is still exposed when a small level of noise is applied to the data. Moreover, MoBES values tend to decrease as more noise is introduced, indicating its potential for use in scenarios that require optimization between utility and privacy.

Building upon these results, our future studies will follow different paths:

1. Temporal evaluation of MoBES, accounting for behavioral drift, which is a tendency when observing users for a longer period of time. In this scenario, evaluations on how the users' behavior changes can provide better information on how to protect them;
2. Explore additional metrics, especially those that could be less sensitive to the protection mechanism, hence, been more useful after the protection or that holds better information, which could possibly be more explored by an adversarial model;
3. Evaluate MoBES under different protection mechanisms (e.g. k-anonymity);
4. Evaluate MoBES under different re-identification techniques (e.g. is it possible to re-identify under protect behavior? how much data is needed to re-identify the users?);
5. Develop an automatic way to select the values of k and τ, simplifying parameter selection.

List of Abreviations

- CDR: Call Detail Record
- DP: Differential Privacy
- Std: Standard Deviation
- Avg: Average
- MoBES: Mobility Behavioral Exposure Score
- GAN: Generative Adversarial Network
- LSTM: Long-Short Term Memory
- SMS: Short Message Service
- RoG: Radius of Gyration
- 2kRoG: 2-Radius of Gyration
- MDIST: Maximum Distance
- NoV: Number of Visits
- AJL: Average Jump Lengths
- SJL: Standard Deviation of Jump Lengths
- NUL: Number of Unique Locations
- ENT: Entropy
- STAT: Stationarity
- REG: Regularity
- DIV: Diversity
- AWT: Average Waiting Time

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- SWT: Standard Deviation of Waiting Time

Declarations

Ethics approval and consent to participate

Data collection was approved by the data owners, and written informed consent has been obtained for all study participants.

Consent for publication

Not applicable.

Availability of data and materials

The Shenzhen CDR dataset is publicly available at https://people.cs.rutgers.edu/dz220/data.html. The Shanghai dataset cannot be shared due to privacy restrictions and non-disclosure agreements, as it contains sensitive personal information. To ensure transparency, the code used in this study is available at https://gitlab.inria.fr/mobility-framework/behavioral-exposure-models/mobes. For dataset access inquiries, contact Aline Viana at aline.viana@inria.fr.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

L.G.S.F: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. N.A. - Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. A.J.K. - Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. A.C.V. - Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. J.M.A.: Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. All authors reviewed the manuscript.

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